STAR WOLF

The Emperor is dead, betrayed by the traitor lords.

You are the Captain, revived from stasis in a secret spacedock.

You are deep behind enemy lines; nowhere is safe.

You command a single ship.

Carry out the Emperor's final edict: revenge on the traitors.

CONCEPT

In Star Wolf, you are in command of a starship. You go to sectors of space, where you will hunt down and destroy the Enemy Flagship.

The game is a top-down 2D real-time, tactical, space combat experience. The novelty of the game is that it is played entirely through a control panel interface. Apart from very select moments, there is no 3D visual of the outside.

Starship captain interact with the world through various ship systems which include navigation, weapons, troops, sensors, and signals intelligence.

The fun of the game is found in:

- managing the position of your spaceship
- Using Sensors to scan enemy ships for weakness and gathering complete data
- Using that data to disguise yourself as an enemy ship, hiding in gas clouds or behind asteroids
- Managing your power to getting into range and angle of fire
- Getting the perfect shot or laying the perfect trap

REFERENCES



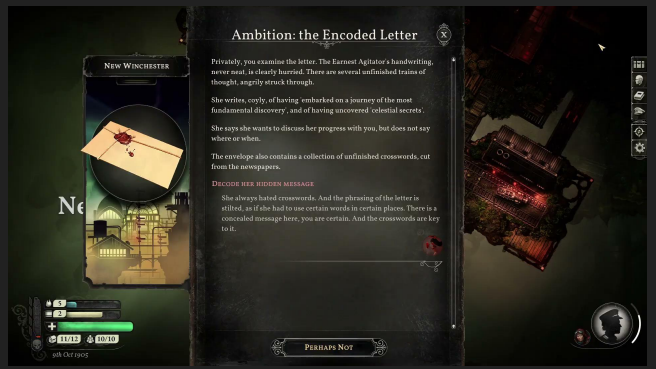
Highfleet - Mechanical UI



In Other Waters - Sonar Style Interface



Battlefleet Gothic - Combat



Sunless Skies - Movement

Scope

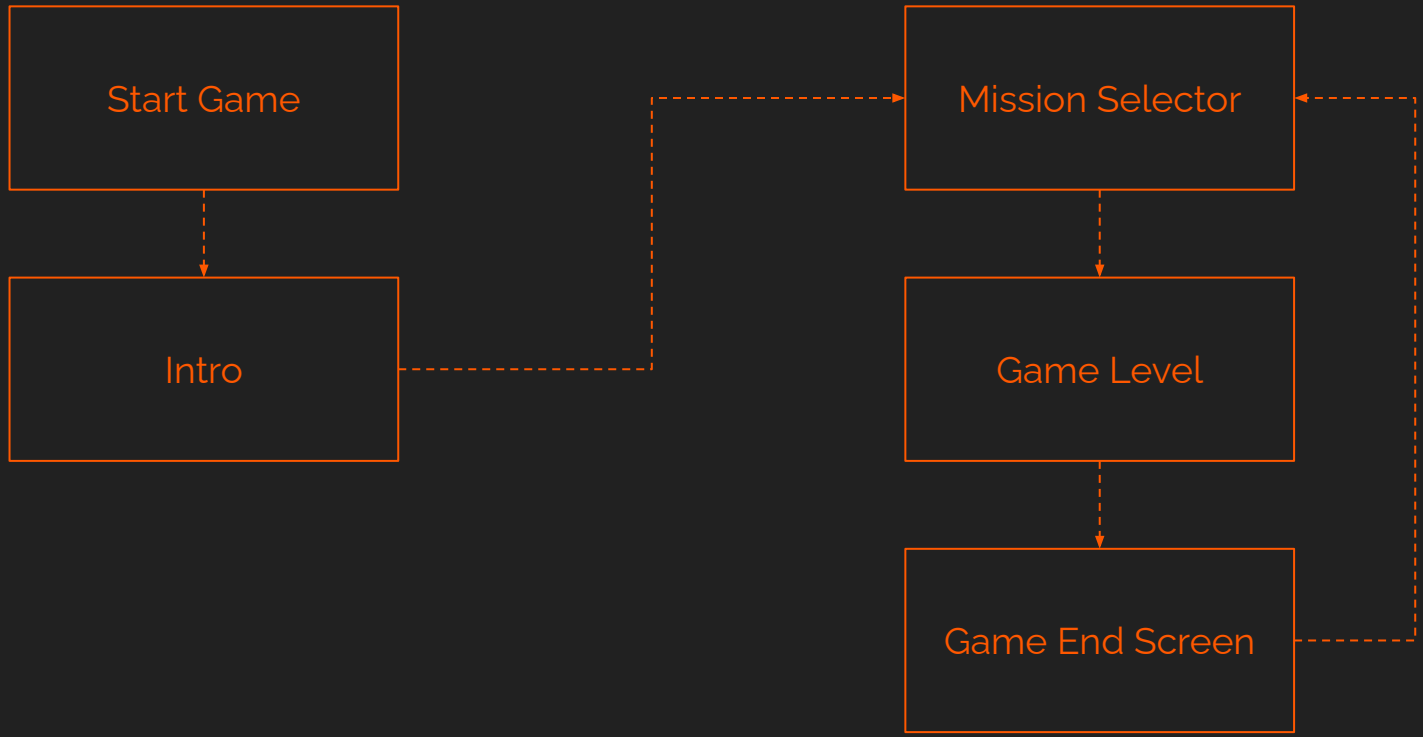
We are aiming to produce one complete mission.

The player begins at the **title screen** and selects the only available Traitor from a card-based interface. This launches the mission. The player enters hyperspace, arrives in the **game level** and must locate, defeat the Enemy Flagship and then escape. Once complete, they return to mission screen followed by a **credits** screen.

This slice must have the following features:

1. A working title screen and credit screen to frame the play session.
2. Interacting with key systems:
    a. Movement
    b. Weapons
    c. Scanner
    d. Abilities
3. Engaging in combat
4. A level layout with meaningful environmental elements and tactical choices.
5. A quest system that checks the game state:
    a. Primary Quest: locate target, defeat target, escape
    b. Discoverable Quest
6. Basic enemy AI that responds to the player's presence and actions.
    a. AI must be able to patrol, investigate disturbances, engage in combat, or retreat.
7. UI layout v1
8. Sound Effects v1

GAME FLOW


Start Game
Intro
Mission Selector
Game Level
Game End Screen

GAME START

When the player starts the game, they are immediately presented with the control panel interface, a screen similar to the main in-game panel.

On screen, a sequence of **Introductory narrative prompts** appear, which tell the story of the game.

Once this message is complete, the **Mission Selection** loads.

A set of cards appears. Each card represents one of the traitors. In the vertical slice, only one card is selectable. Clicking the card expands it, revealing flavour text that describes the selected Lord and their role in the betrayal of the Emperor. The player presses the Go button to begin the mission.

Immediately after, a pre-launch sequence begins. The control panel displays a rapid stream of sci-fi status updates - “Initializing drives,” “Unclamping,” “Loading ammo hoppers,” etc. - to convey the Star Wolf coming online.

The player is prompted to enter a short arrow key sequence (e.g. up, down, up, down, left, right) to initialize the hyperdrive. Once entered, a wireframe visual of the wormhole appears, followed by confirmation: “Hyperdrive Initialized.”

This sequence is used to build atmosphere, audio-visual immersion, and buy time for the game to load the level in the background. After a few moments of launch audio and system sounds, the player is loaded into the level.

GAME LEVEL

Each mission takes place in a large, handcrafted region of space. The level is has random shape and is filled with structure and detail. There is no true deep space. Every mission has a unique environmental identity: the rings of a shattered moon, the surface of a gas giant, the wreckage fields of a dead fleet, or the structure of a broken starbase. These backdrops are functional. They affect radar returns, scanner noise, and tactical movement.

When the player arrives, they appear on one edge of the map. They begin with minimal, outdated data - old Imperial records that show at key landmarks: a listening post, an abandoned dock, a comms tower. These are unverified and must be scanned or approached to confirm. The first phase of the mission is exploration. There is no fuel mechanic.

All enemy ships are passive at the start. The *Star Wolf* is broadcasting a false ID, and enemies see it as a friendly or neutral contact. Hostile units patrol, dock, or remain stationary until provoked - by proximity, weapons use, or failed deception.

The player's job is to untangle the level. Scan signals. Track patrols. Identify weak points. Strike when the moment is right. The boss—the Traitor Lord's vessel—is hidden within the level, surrounded by forces and misdirection. The player must determine where it is, how it's protected, and how best to kill it.

Once the Lord is dead, the extraction vector appears. It is always at a far edge of the map. The player must escape, evading any remaining enemies or hazards. There is no cutscene. The mission ends when you reach the edge and disappear into the void.

GAME OVER

After completing the level, the player is shown a quick end screen which runs the credits.

Then they are returned to the mission screen and receives a small narrative blurb talking about their victory.

The player may once again select the mission to replay it, or close the game.

Order of Operations

Explore the world

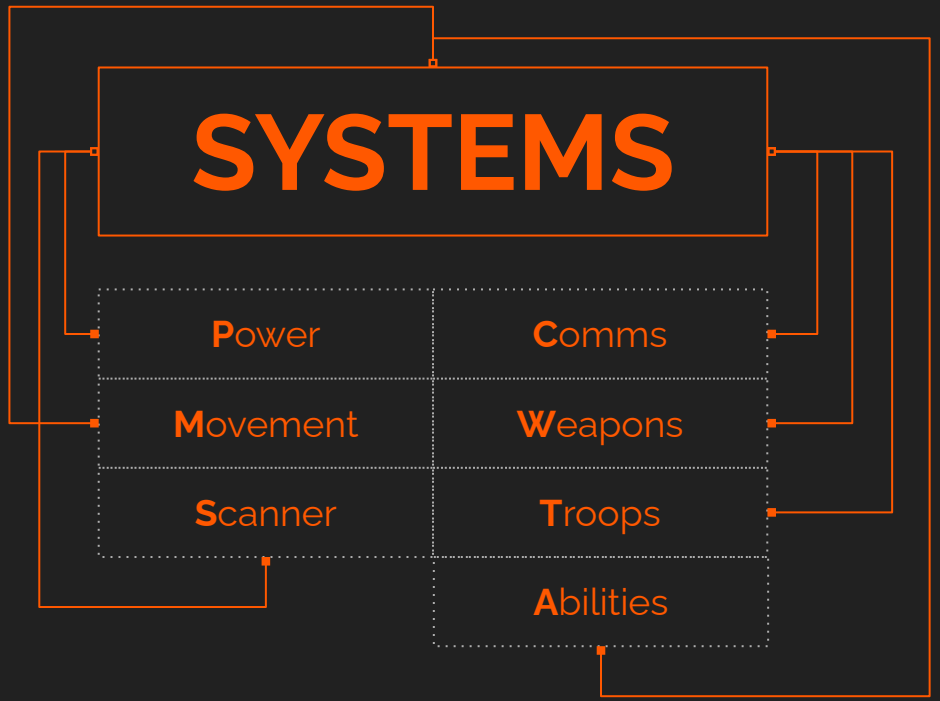
- **Move** around
- Find the enemy
- Find points of interests
- Identify correct location to setup strike

Use the **Radar** to figure out what's going on in the world and identify situations; use that to make good decisions (WHAT TO TARGET)

Use **Weapons & Systems** to engage

Calibrate **Power** to optimise your approach

Take care of yourself while doing so

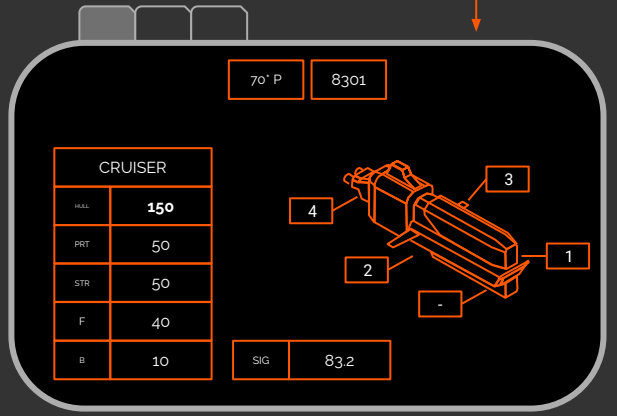

SYSTEMS
Power
Comms
Movement
Weapons
Scanner
Troops
Abilities



World Map

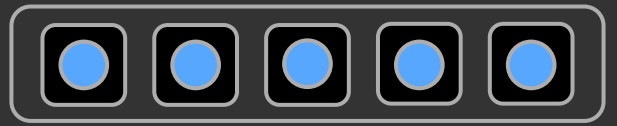
Movement Interface (Currently Sonar)

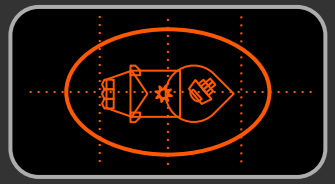
Systems Screen





Power





Ship Health



Readouts

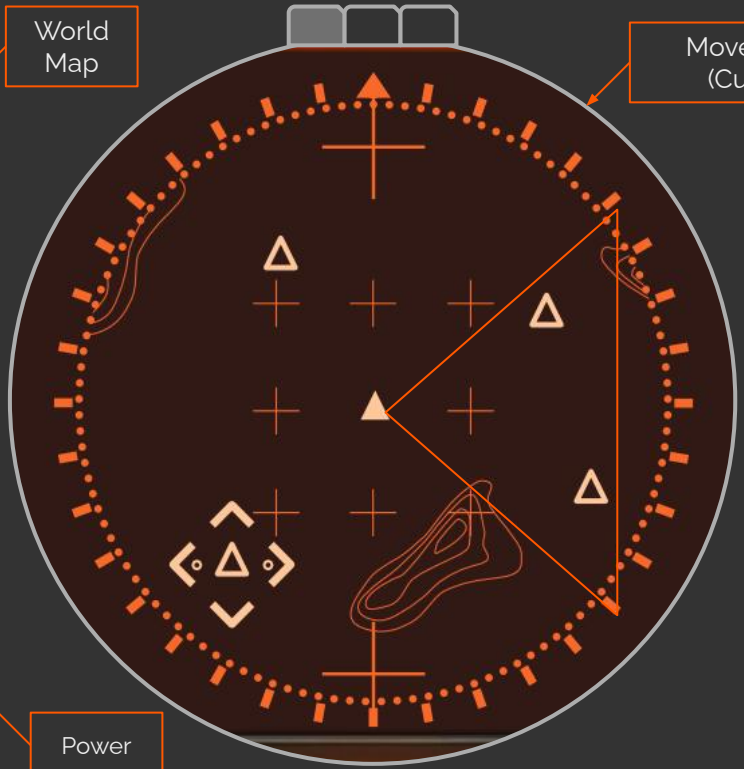
Abilities

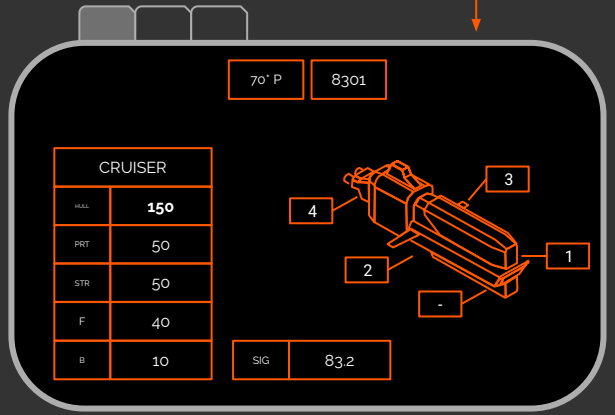


World Map

Movement Interface (Currently Sonar)

Systems Screen

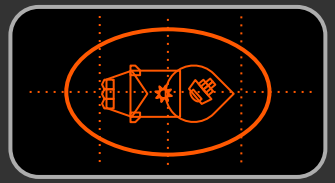




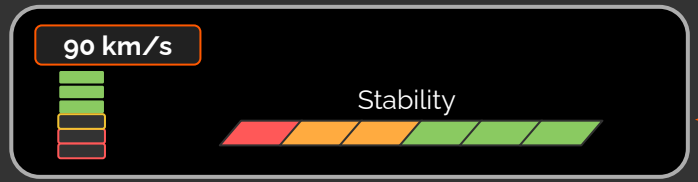


Power





Ship Health

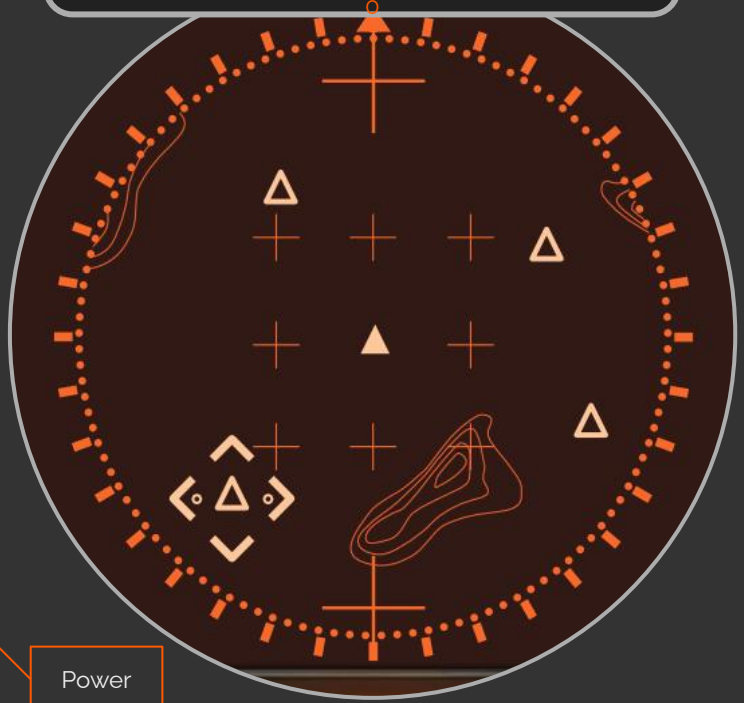


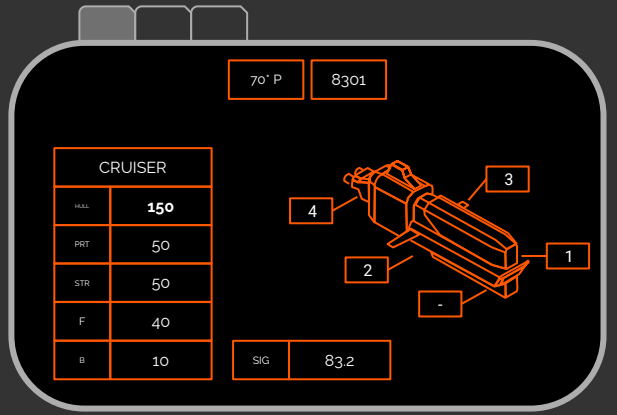
Readouts

Abilities



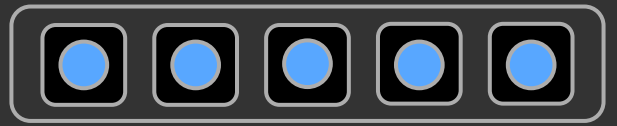


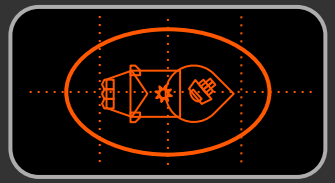




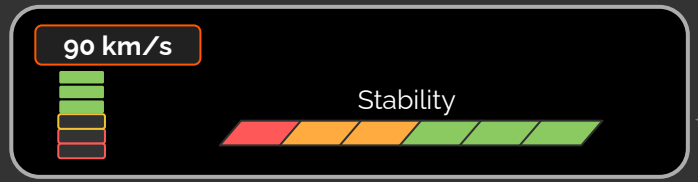


Power



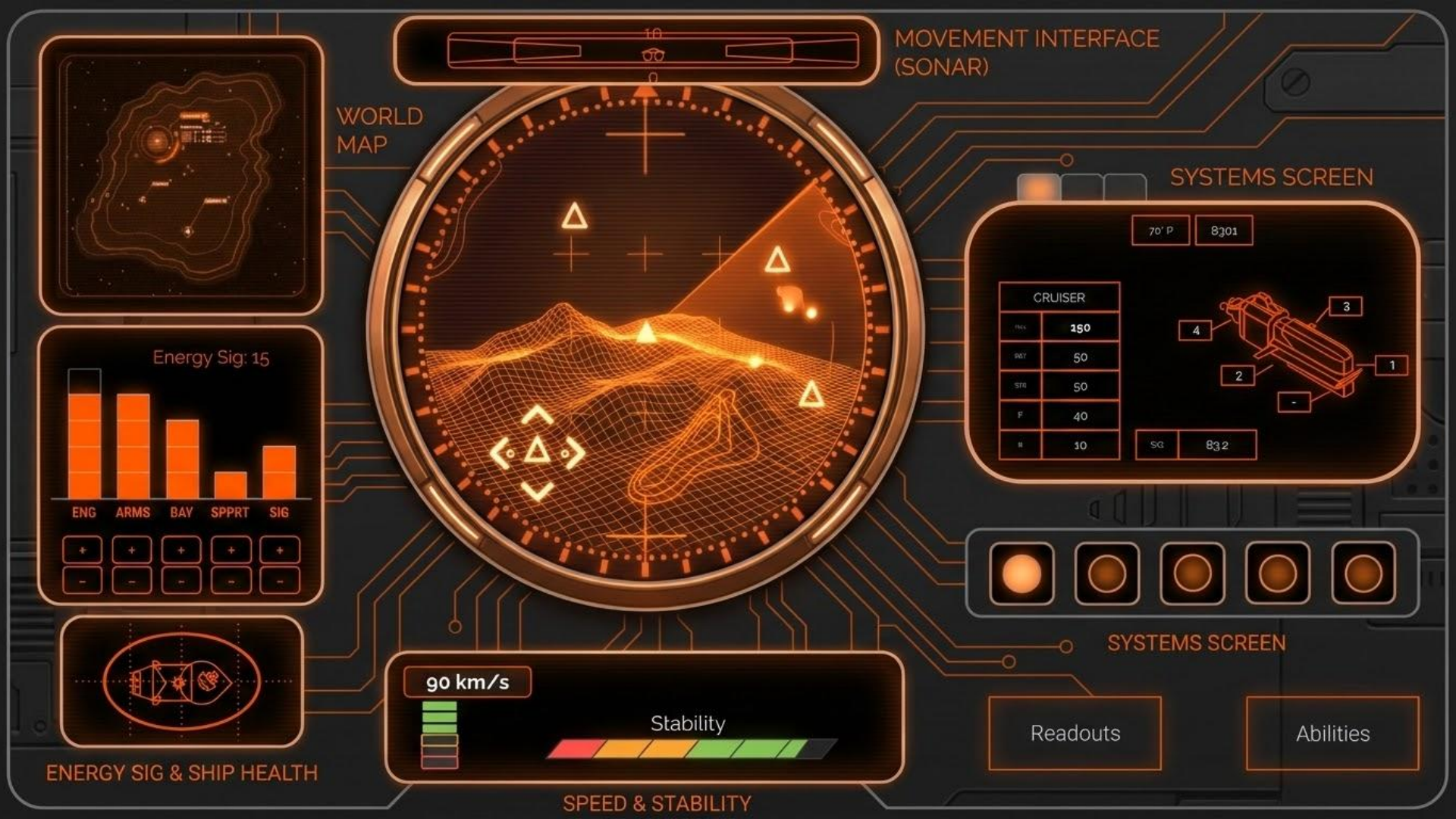


Ship Health



Readouts

Abilities

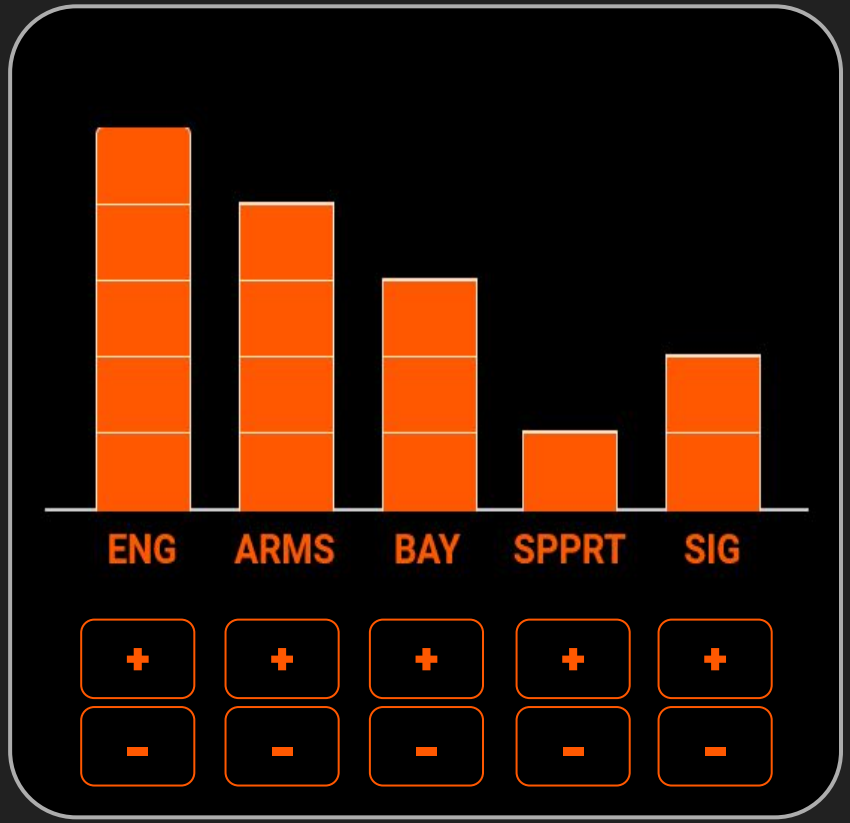

MOVEMENT INTERFACE
(SONAR)
WORLD
MAP
SYSTEMS SCREEN
70' P
8301
CRUISER
150
50
50
40
10
SG
832
3
4
1
2
Energy Sig: 15
ENG
ARMS
BAY
SPPRT
SIG
SYSTEMS SCREEN
90 km/s
Stability
Readouts
Abilities
ENERGY SIG & SHIP HEALTH
SPEED & STABILITY

POWER

- Power is at the heart of all systems in Star Wolf
- Power comes from the Reactor, which is a capped wallet
- All systems needs at least **1 power** to function
- Sending additional power to a system unlocks bonuses

Bonuses

Engines	Acceleration is increased by 10% per PWR - 1
	MAX PWR unlocks Supercruise which 3x Top Speed
	Stability decacy per turn reduced by 5% per PWR - 1
Sensors	Scanner: Perfect hit windows increased by 1% per PWR - 1
	Comms: Intercept time increases by 1 second per PWR - 1
Weapons	Cannon load time reduces by 2.5% per PWR - 1
	Boarding pod ranged increased 10% per PWR - 1
Support	Ability cooldowns reduced by 5% per PWR - 1



POWER

In the power panel, there is a switch corresponding to each system. Switches have 3 settings: Draw, Standby, Vent.

Switches start on Standby. Pressing on the Switch will set it to DRAW. Power will begin to fill one bar at time, drawing from the reactor. Each bar takes around 2 seconds to fill. There is no visual of a power bar gradually. Ever 2 seconds, a new bar will appear full. The filling is done off-screen.

Press the switch again to set it to STANDBY. This is will stop drawing power from the reactor once the current bar is filled or the reactor runs out.

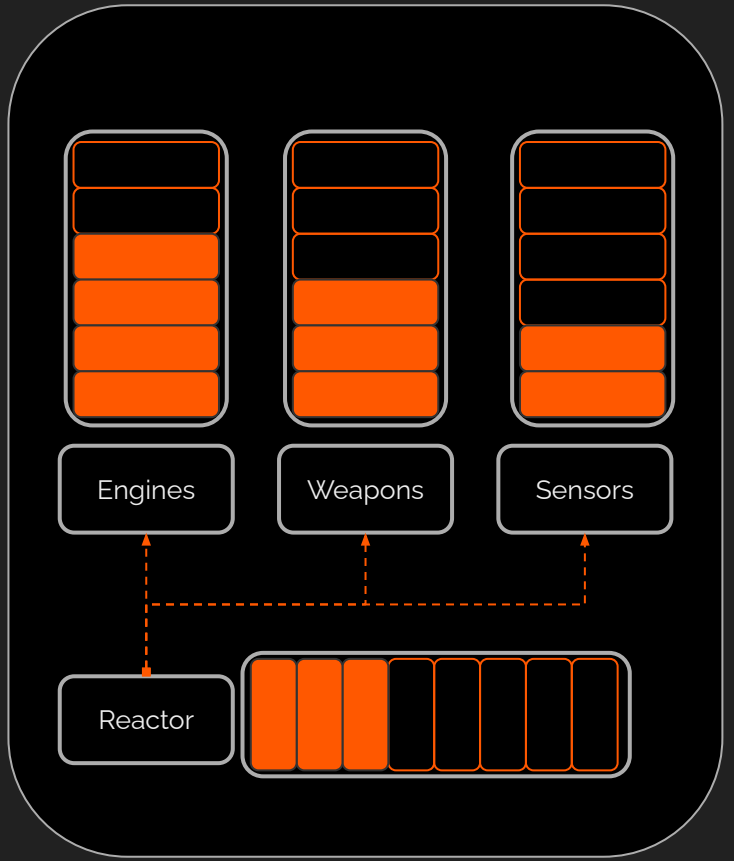
Power is routed evenly so if you press the switch to Engines then the engine bars will star filling at a certain speed. If you then start also routing power to Weapons, then the fill rate per bar will reduce by 50%; this applies evenly across all active functions, which means that activating all functions at once will make all of them draining at 20% of their speed.

Bars are whole numbers, so a system cannot have 4.3 power. This would round down to 4.

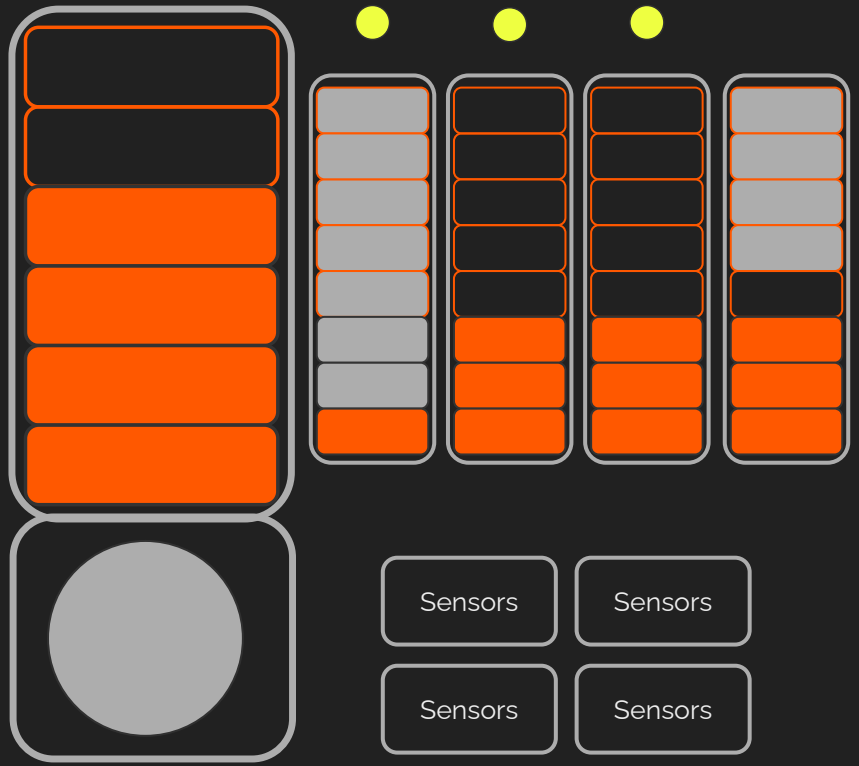
Pressing the switch again will now initiate a Vent. This flow is fixed:
Standby -> Draw -> Standby -> Vent -> Standby

VENT will begin to deplete power at a rate of 1 bar/second until it is 0. You cannot stop venting once triggered. The power will leave until the system is 0.

The reactor bars will fill as each bar in a system depletes. You can vent multiple systems at a time.



POWER



DETECTION

It is important to remember that your ship is a destroyer that relies more on **stealth** and **patience** than a straight up fight.

So while you can route all the power in the reactor to various systems, remember than you will be more noticeable on enemy radar.

So some basic principles:

Power in the reactor cannot be detected. Assume that the reactor is well shielded and insulated,
Power in systems can be detected. Using power (flying at speed, firing weapons) adds to the **Energy Signature**. Just as an enemy's energy signature will appear on your radar, so will yours appear on theirs.

Managing your energy signature is important to maintain the element of surprise. Later, we will read about scanning. Getting data on other ships let's you adjust your energy output to match theirs, which will let you get in close and maybe avoid fighting until you're ready. Similarly, you can power down all systems and hiding in gas clouds and debris fields.

Detection can be influenced by a variety of environment elements which we will talk about later.

Generally, understand that Detection = Sum of Energy in Systems + Active Ability + Environment Effect

So having 3 energy in weapons + firing the gun (+ energy) + Gas Cloud (energy reduction %) would be:
(3 + 2) - 20% X (3 + 2) = 5 - 1 = **4** (rounding up).

MOVEMENT

The ship moves in real time on a flat 2D plane. There is no verticality no teleporting, snapping, or instant turning. Movement is based on speed, thrust, and turning input. Momentum build up and slow downs are both gradual.

Flying the ship is very simple:

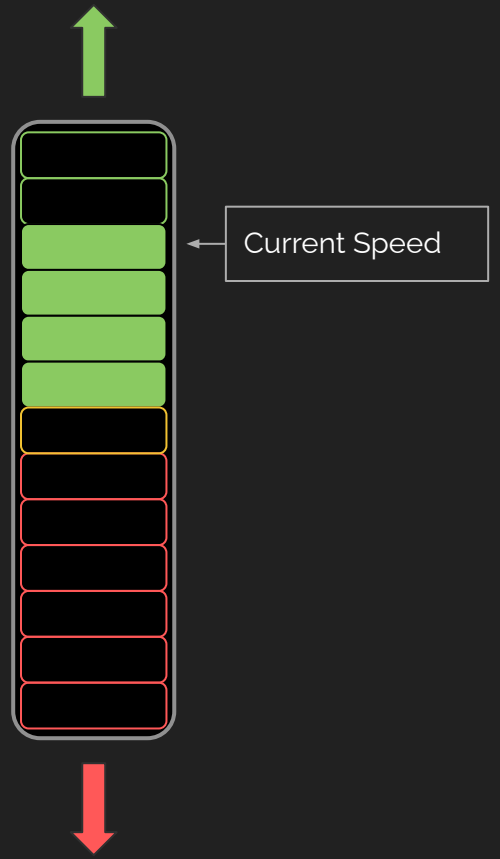
- **W** / **S** to adjust thrust
- **A** / **D** to turn the ship
- **Q** / **E** to perform an emergency dodge to the left or right

When you press W, you increase forward thrust. Eg. If you set thrust to 40, the ship accelerates up to a forward speed of 100. If you then pull back to -50, the ship will first decelerate to 0, then begin reversing at speed -50. There is no instant braking. All motion changes are gradual. Each bar = “+20” units of speed.

We have an editable **Rate of Acceleration**

Ships have a **Turn Rate** of a predefined number of degrees (with 1 decimal place) per second. This is also adjustable in Inspector.

The ship can spin unexpectedly if you mismanage turn and thrust - but that’s not always bad. Quick rotations at the cost of stability can be tactically useful to get weapons on target or escape a bad heading.



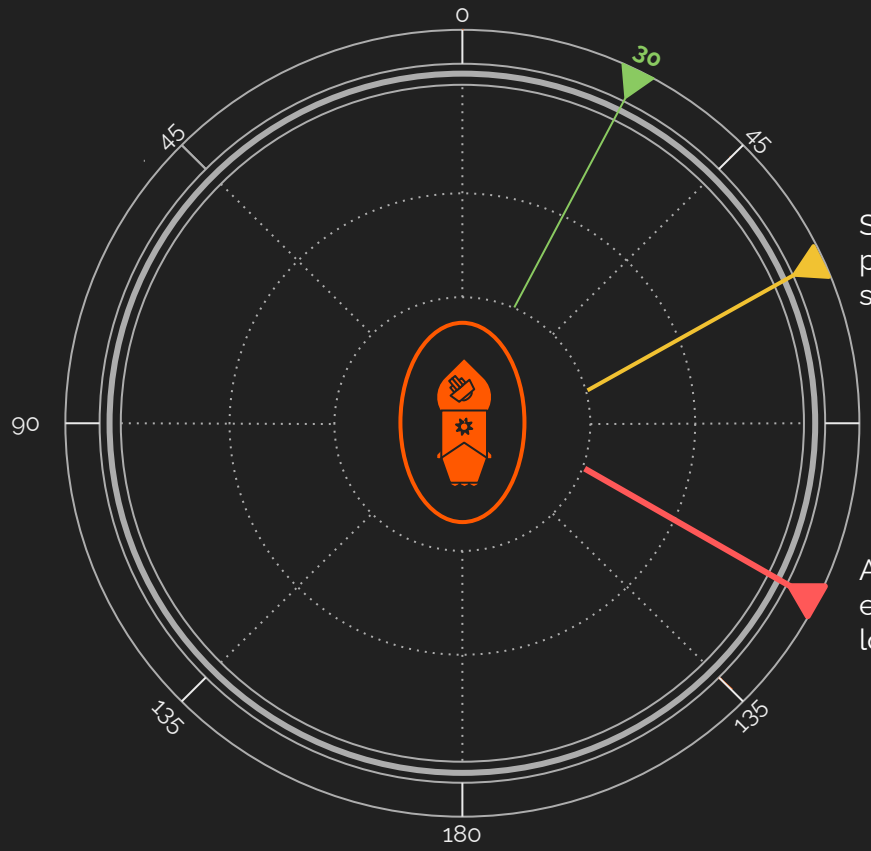
MOVEMENT

Stability

The ship has a Stability stat. This tracks how safely the ship can manoeuvre. Turning at low speeds drains very little stability. Turning sharply at high speed will rapidly reduce it. When stability is depleted, the ship will incur internal damage to the hull and systems.

There are three levels of turn severity:

- **Green Zone**: Small heading changes within 60˚. Low risk.
- **Yellow Zone**: Moderate turns (~45˚ arcs). Higher stability drain
- **Red Zone**: Large or fast turns (~90˚ or more). Extreme stability drain.



Sharp Turn; needs more power. Greater load on stability

Aggressive Turns; Needs even more power, extreme load on **Stability**

MOVEMENT

Dodge

Sometimes, you will need to avoid something in a pinch. This is when dodge comes in handy.

Press **Q** to left and **E** to go right.

This is not a turn. The whole ship just shifts to the side.

Dodge has a huge impact on **Stability**.
At best, you can dodge twice in quick succession. The cost in Stability would be 45% in both cases, leaving you at 10% and most likely to incur severe damage if you manoeuvre immediately and aggressively after.



MOVEMENT

Interface

The navigation interface in *Starwolf* has two core sensor modes: **Radar** and **Sonar**. These are visual camera feeds meant to mimic instrument panels. These are UI-rendered 1980's styled representations of internal displays. The player will keep toggling between these 2 settings.

Both views appear in the **navigation panel**. The ship icon remains fixed at the centre of the screen, and points toward the top. The world rotates around it based on heading. Movement, orientation, and threats are all interpreted through these two modes. Players can toggle between both.

Radar view puts the player character at the center. The world orients around the player and moves based on player inputs.

It displays long-range information. Objects in this view are rendered as fixed-size abstract shapes (circles, triangles, crosses) and do not scale i nsize based on distance. These shapes represent known contacts: ships, stations, asteroids, or unknowns. Each contact's position is calculated relative to the player's transform and placed accordingly on the map. Radar is used for strategic navigation - plotting routes, watching patrol paths, and identifying major structures. It does not provide detail or visual fidelity. Radar **updates every X seconds**, so players will see blips teleporting between pulses. **It covers a radius of 25,000 units (editable in the Inspector).** We want to be able to edit **Update Interval** and **Radius**.

Sonar covers the near field, with a radius of 5000 units (also editable). It shows a more detailed view of the local space, rendering objects in wireframe silhouettes to scale with the ship. These are pre-made 2D PNGs. It is used for precise navigation through dense areas, like debris fields, broken hulls, or complex structures. Think of Sonar as a to-scale version of the world. It updates in realtime. If something enters your sonar unscanned, you will have to scan them to get information.

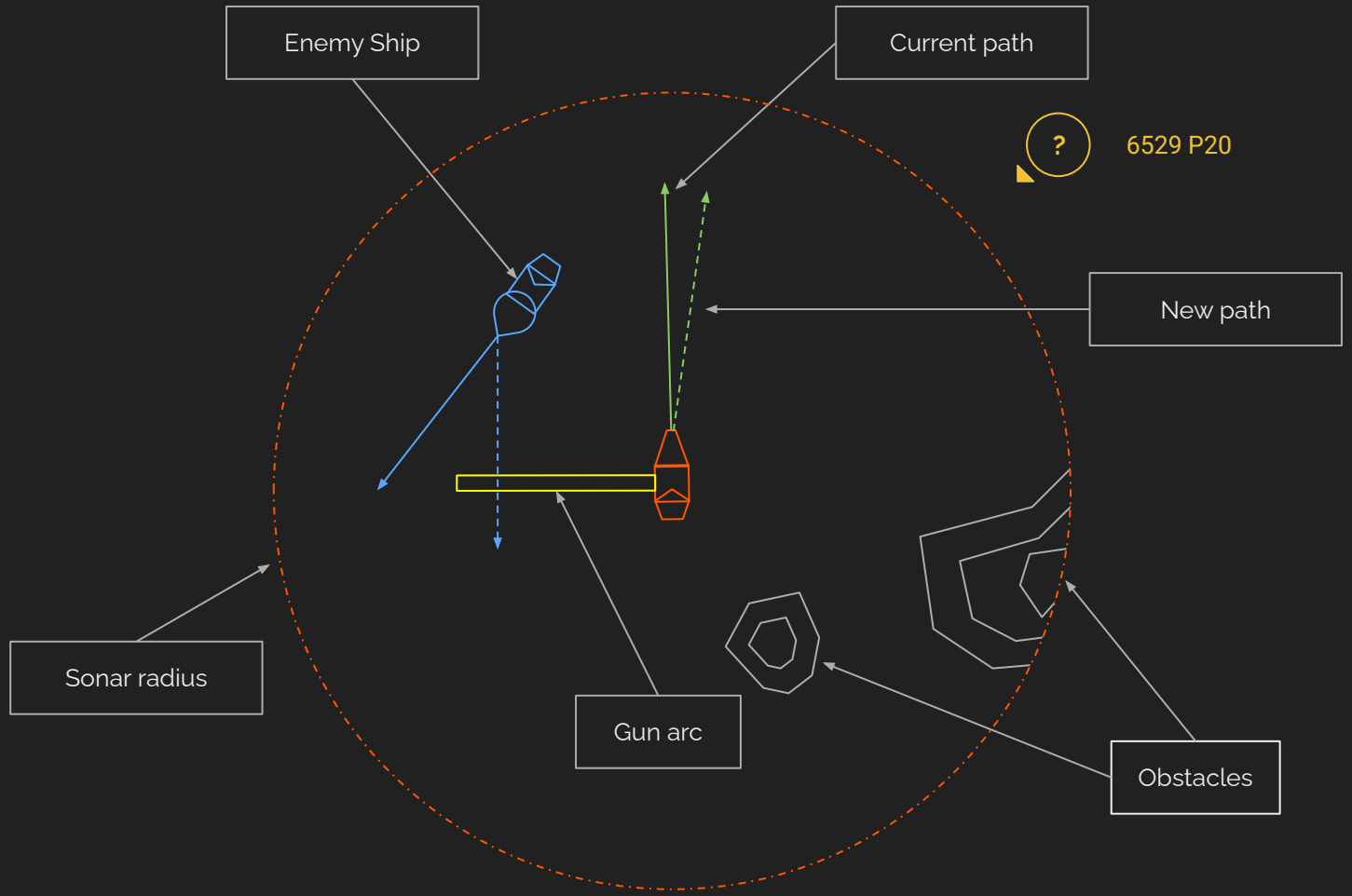
MOVEMENT

Radar should be implemented using Unity cameras. These are not scene renderers. They are pure UI systems designed to simulate shipboard instruments. In earlier prototypes, using cameras led to significant issues -warping, scale issues, layering problems, and provided no real benefit. It is important to note that the scale of the radar and sonar are not 1:1. A radar will show a signal with some data. Scanning the radar signal will give you a sense of what's over there. Toggling to Sonar will show you the scale of the object in relation to your ship.

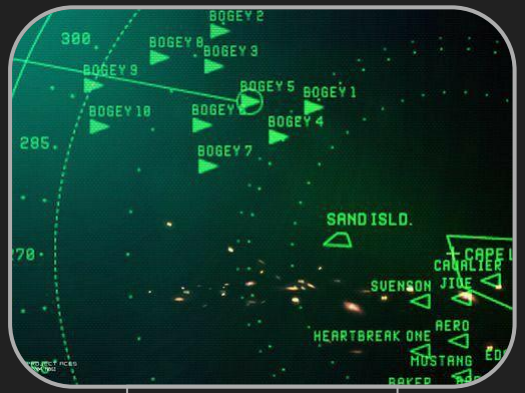
All sensor visuals should operate entirely in 2D space (XY plane). We can have an array of objects which is responsible for tracking all detectable objects on Radar, converting their world positions into local UI coordinates, and culling anything that falls outside the sensor's range; and instantiates them when they are in sonar range.

Objects (whether blips, icons, or wireframe silhouettes) should be drawn using static UI sprites. Effects like CRT distortion, bloom, scanlines, or flicker will be layered in using shaders.

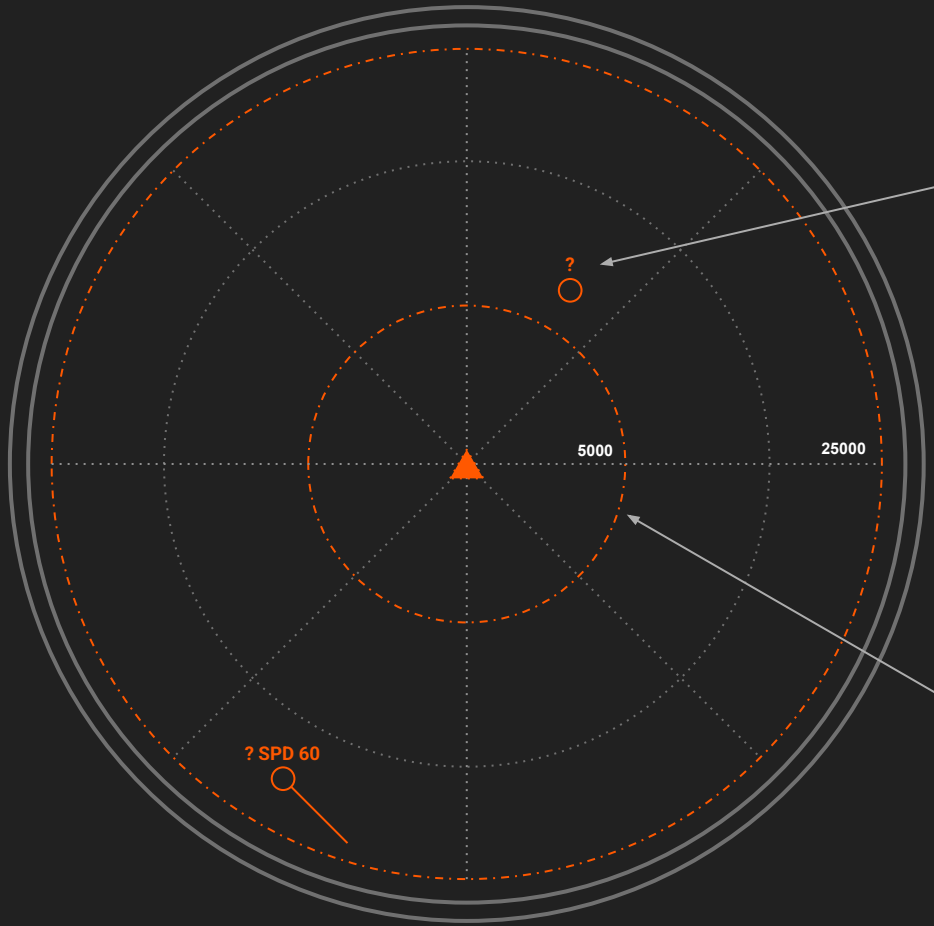
Sonar


Enemy Ship
Current path
?
6529 P20
New path
Sonar radius
Gun arc
Obstacles

RADAR



Reference

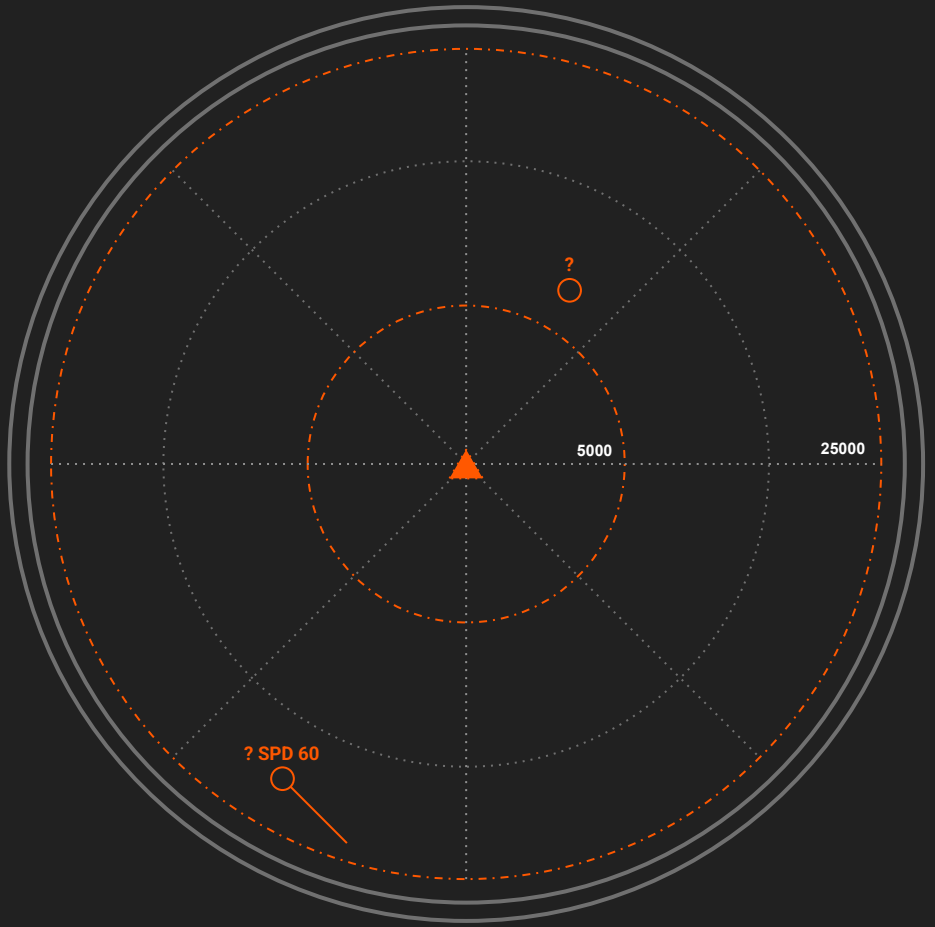


Scannable OBJs

Sonar Radius

Scale Mock





INTERACTION SCREEN

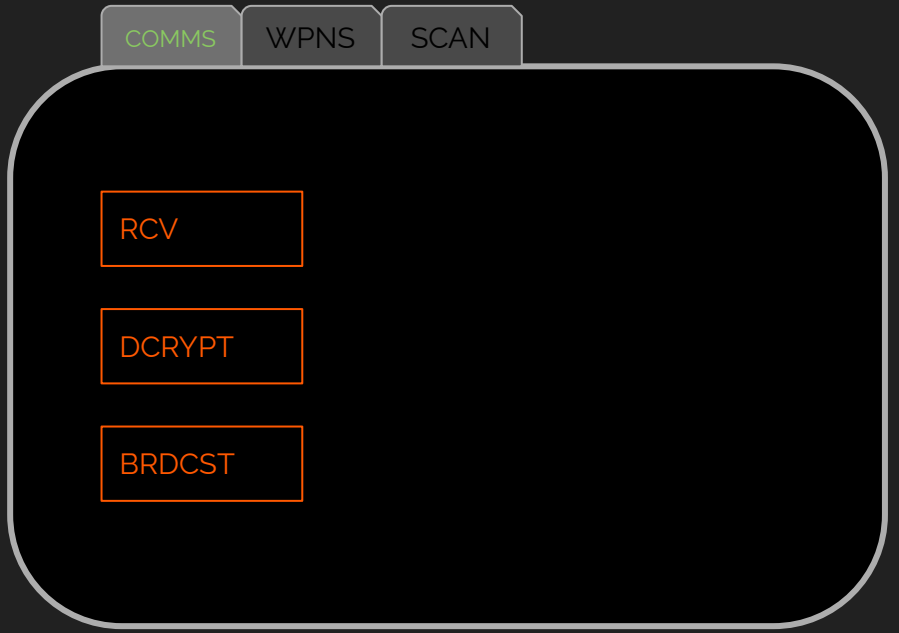
This is screen is where all ships systems apart from Navigation and Power can be interacted with and is second most crucial element on the screen,

Players can toggle between ship systems. This screen will show:

- Scanner
- Weapons
- Comms

Sub-systems are nested within these panels.

We will use flashing lights to show what's pending.



SCANNER

The Scanner is used to gather detailed information about objects detected on the radar. It reveals data such as ship type, hull integrity, power output, and current signal profile.

When the player toggles to the Scanner pane, they may select any target currently detected on the radar. Once a target is selected, the Scanner mini-game begins.

Feeding power to the Scanner increases the size of hit windows, making the mini-game easier and more forgiving.

Larger or more advanced targets feature more pulses. Some targets may require multiple scan attempts to retrieve full intel, especially when scanning at low power.

SCANNER

A horizontal slider scrolls from left to right.

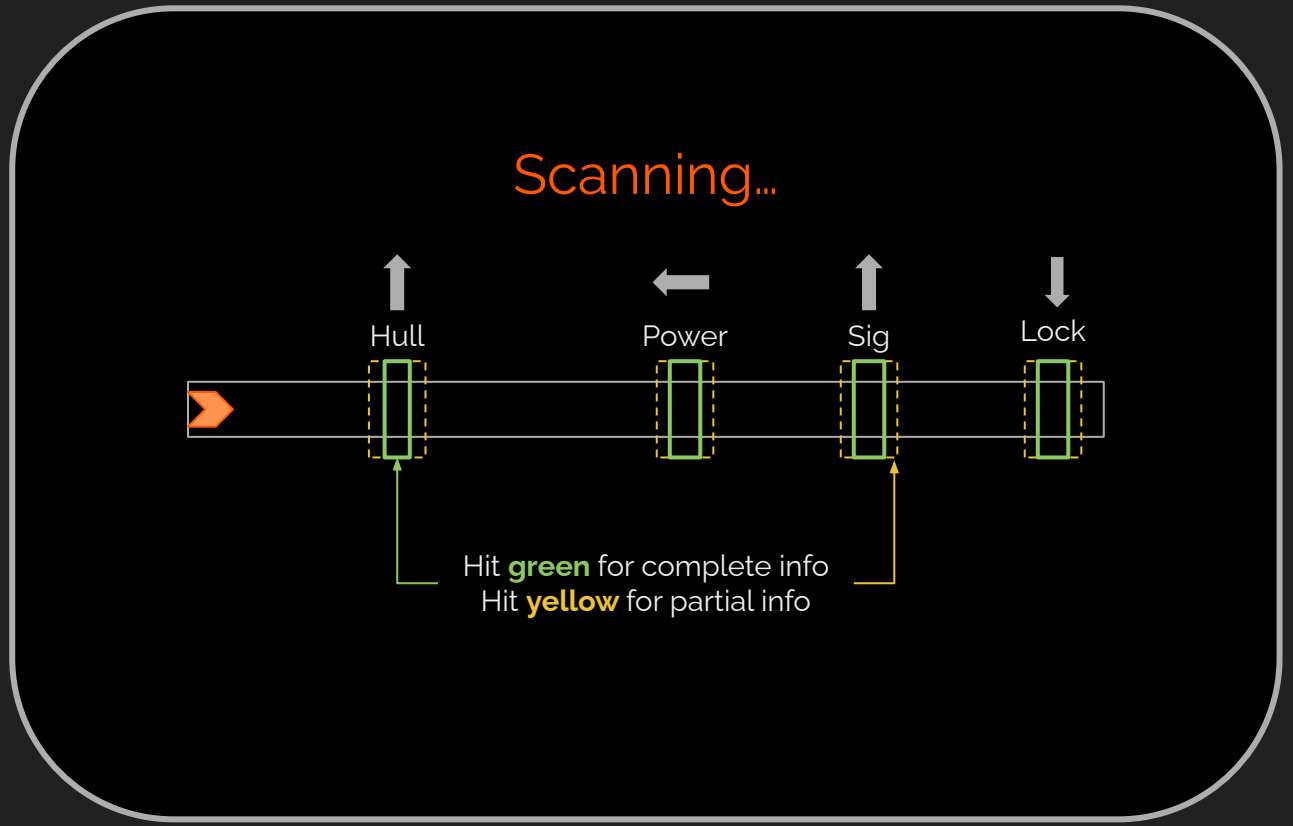
A sequence of bars appears on the track.

The player must press a set of arrow keys at the right moment as the scanner line crosses each bar.

Each successful hit reveals a layer of data. Misses result in incomplete or partial intel. Think of it like a Guitar Hero track, without music.

A win results in a lock which allows a player to track the ship on the world map for sometime (in minutes)

Rescanning the same object (to gather any missing data) has a cooldown period of 30 seconds. This can be changed with Power.



SCANNER

Direction

70° P

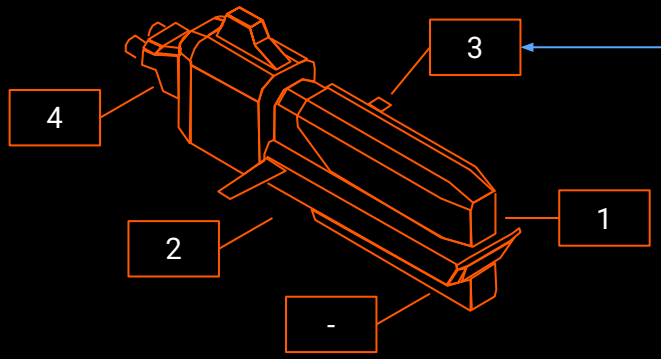
8301

Distance

CRUISER	
HULL	**150**
PRT	50
STR	50
FRNT	40
BCK	10

Hull Total

Hull health breakup



Power output for each system

SIG 83.2

Signal used to identify the ship

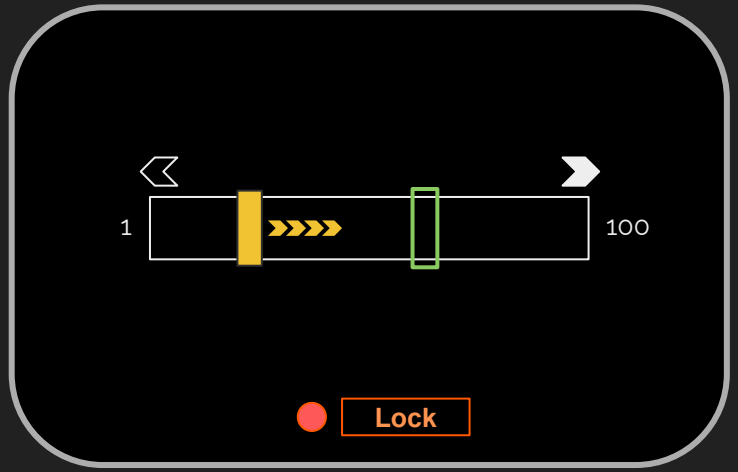
COMMS

Captured signals yield one of two outcomes:

- Readable transmission: displayed immediately in the log.
- Encrypted message: marked as ciphered, requiring a separate minigame to decode later.

You can only attempt to lock once per signal. A miss discards the signal.

Signals will always come from within a range of **200k** units (editable) so a player will be able to make some assumptions and do math around the maximum possible search radius.




```
==|RAJA|==

8 Hrs to Delta Point
Speed 90
Initiating SUPERCRUISE

==|ASUR|==
```

COMMS

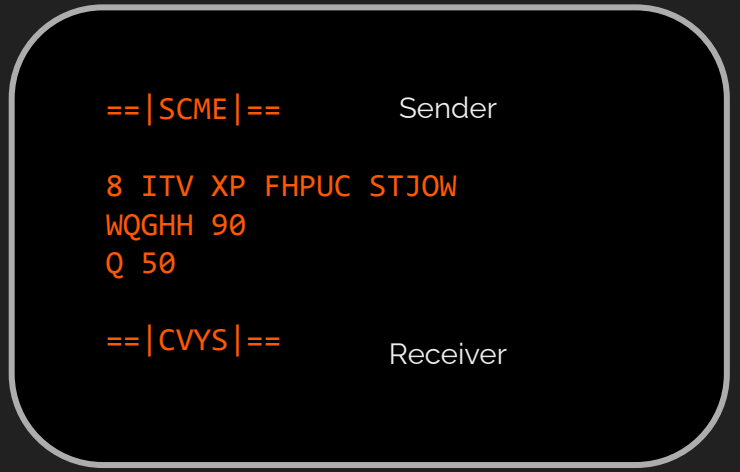
Decrypting messages

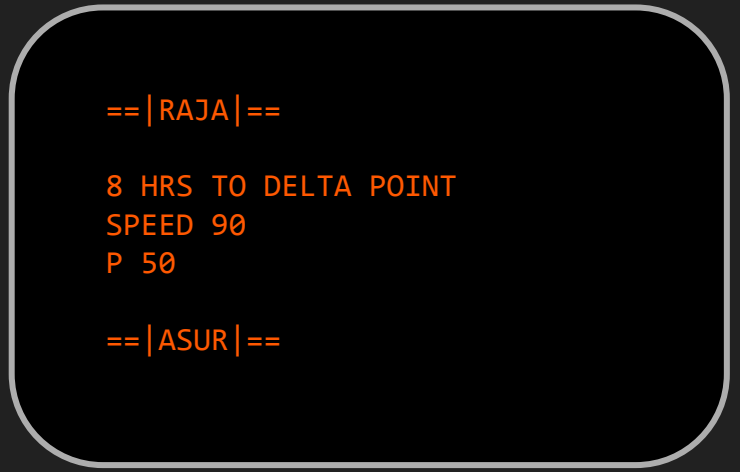
Some intercepted messages will be in code and display as garbled text. To decode them, the player uses the *Decrypt* option in the Comms system. This interface contains four cipher wheels.

Rotating each wheel shifts specific letters in the message up or down the alphabet. Each wheel has a number between 0 to 25.

- The encrypted message is made of letters A–Z.
- Each wheel controls every fourth letter of the message:
  - Wheel 1 affects positions 1, 5, 9, 13, etc.
  - Wheel 2 affects 2, 6, 10, 14, etc.
  - Wheel 3 affects 3, 7, 11, 15, etc.
  - Wheel 4 affects 4, 8, 12, 16, etc.
- As you turn a wheel, the linked letters switch in real time.
- Linked letters are show in red so the player knows which ones will change.

You rotate each wheel until the message begins to make sense.





COMMS

	S	C	M	E	8	I	T	V	X	P	F	H	P	U	C	S	T	J	O	W	W	Q	G	H	H	9	0	Q	5	0	C	V	Y	S
W1	1					5				9				13				17				21						25						29
W2		2					6				10				14				18				22								26			
W3			3					7				11				15				19				23								27		
W4				4					8				12				16				20				24								28	
	R	**A**	**J**	**A**	**8**	**H**	**R**	**S**	**T**	**O**	**D**	**E**	**L**	**T**	**A**	**P**	**O**	**I**	**N**	**T**	**S**	**P**	**E**	**E**	**D**	**9**	**0**	**P**	**5**	**0**	**A**	**S**	**U**	**R**

We ignore the numbers 8, 90, 50, and all spaces. Then we rotate the dials left or right to change the letter from S to R for example.
The dials always start at 00 and it wraps around from 25.
Thus by rotating the dials, we will be able to convert the letters to solve the puzzle. All letters associated with a wheel change as one,

COMMS

Over time, the player will build a list of callsigns from less encrypted messages and be able to substitute the letters correctly. If the player already knows known callsigns like ASUR, RAJA, or VEER, they can try adjusting the wheels until the top or bottom lines match a known callsign. Once a match is found, the rest of the message should begin to align.

Enemy ciphers change occasionally. To aid decryption, players can recover cipher fragments from boarding enemy ships. These fragments reveal some or all of the correct values (e.g. “Wheel 4 = 21”) and can be assembled into a full key.

Players do not need to decrypt messages to win. But decryption reveals movement orders, alert status, and sender location -critical tactical intel. Some messages include ETA, heading, or relative bearing to the player, which can be used to reverse-calculate the enemy’s position and type. There is no penalty for guessing incorrectly.



COMMS

Broadcasting

The last piece of Comms is the broadcast function.

This has two pieces:

- A list of scanned ships
- A broadcast numbers (0.0 to 99.9

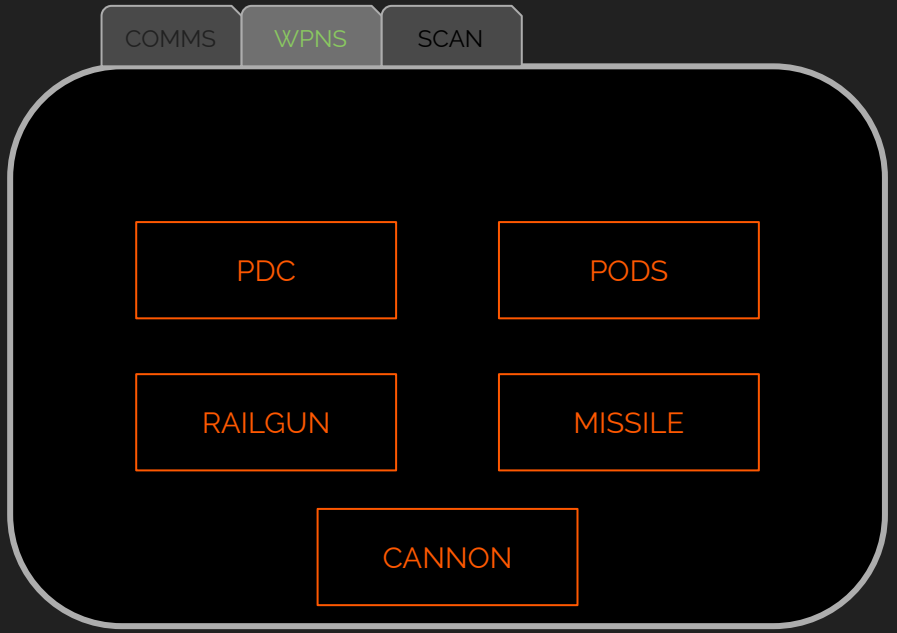
You can view the list and use it to adjust your power readings and speed to “disguise” yourself as a scanned vessel. Similarly, you can set your broadcast frequency to the number of the ship.

WEAPONS

The Star Wolf is equipped with a variety of weapons. Each weapon has an individual PWR cost. You will toggle to Weapons Tab on the Interaction Screen and select the weapon you want to arm.

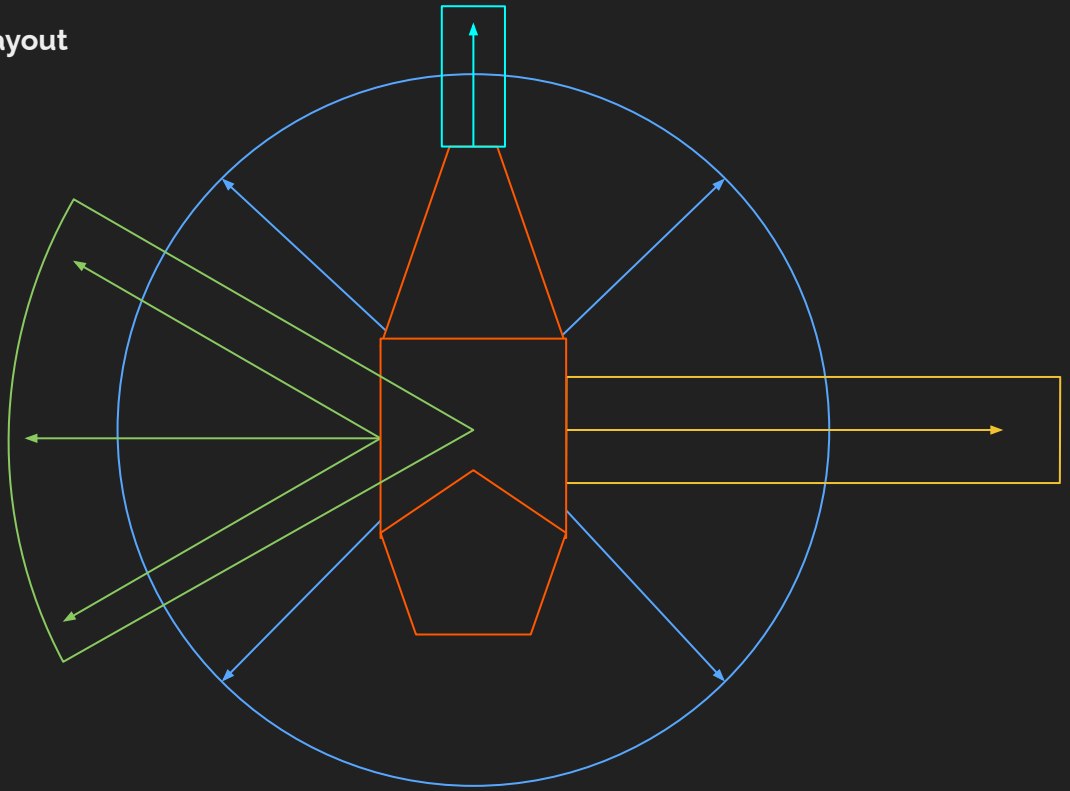
1. **Cannons:** A kinetic weapon which fires shells
2. **Boarding Pods:** Launches pods of space marines to board enemy ships
3. **PDCs:** Rapid-firing guns that shoot down incoming fighters and missiles.
4. **Missiles:** Guided weapons which lock on and track targets.
5. Railguns: Long range sniper weapon, high finesse

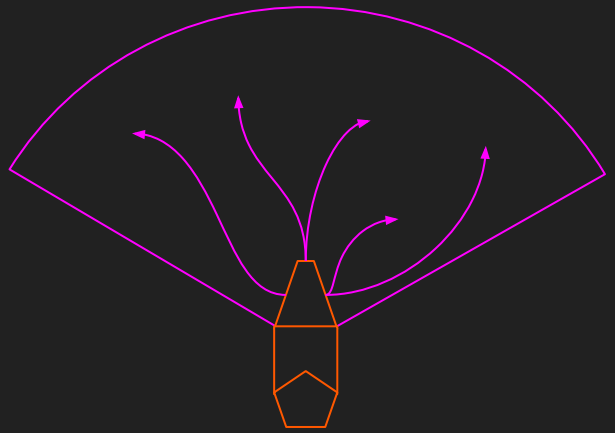
- All weapons have:
  - Range
  - Angle Of Fire
  - Reload Time
  - Ammo Count
  - Damage
  - Dmg Modifier



WEAPONS

Layout





PODs

PDCs

CANNON

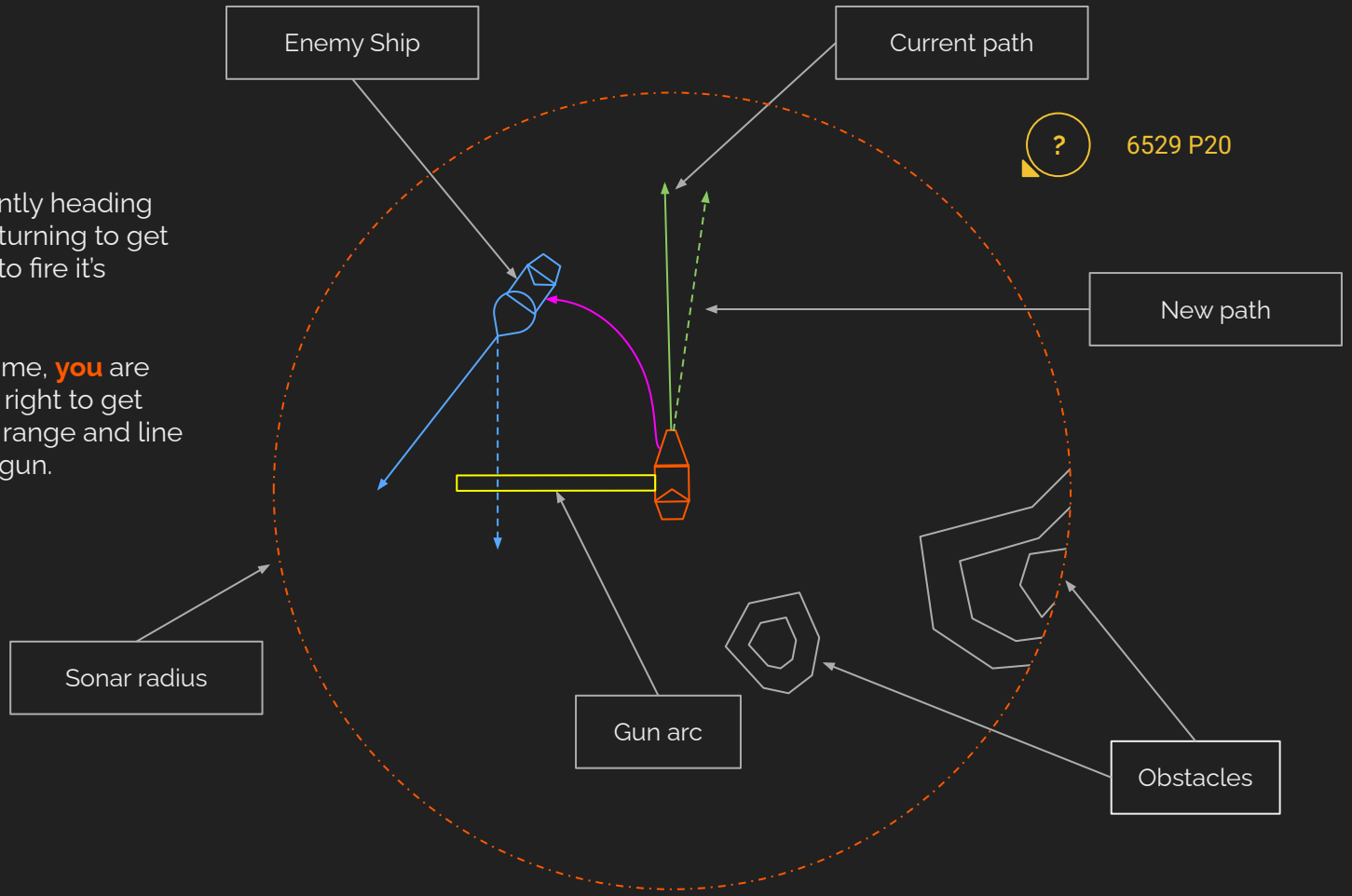
RAILGUN

MISSILE

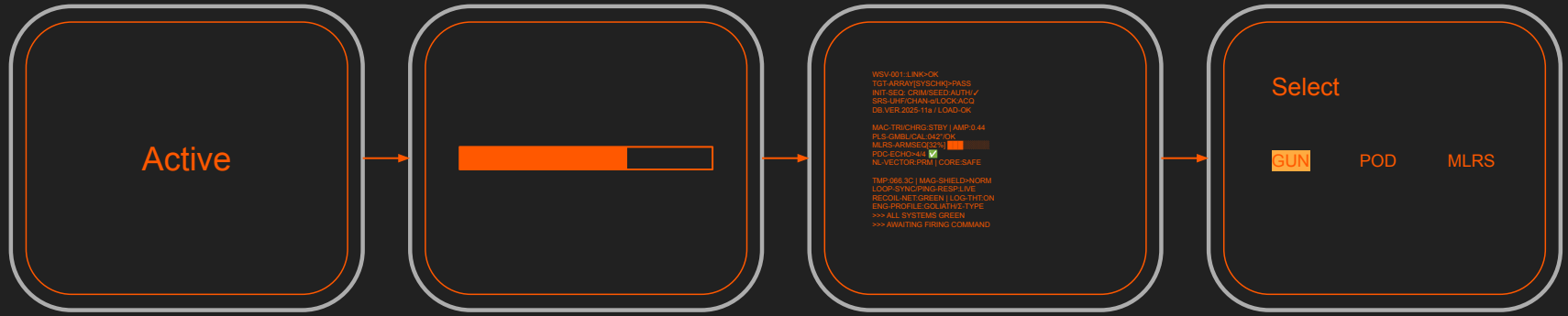
Scenario

BLUE is currently heading past **you** and turning to get into an angle to fire it's weapons.

At the same time, **you** are turning to the right to get them into the range and line of fire of your gun.



Weapon - Flow and Game feel



1. Start by routing power to the Weapon SYS

1. SYS Intializes
2. Screen flicker to life
3. Loading bar
4. Telemetry on screen
5. Select Weapon
6. Open Gun Readout

Broadside Cannons

The **CANNON** (ingame: **Macrocannon**) is a heavy kinetic weapon mounted on both side of the Star Wolf. It fires high-velocity shells across space with a fixed firing angle aligned to the ship's X axis. Each shot travels visibly across the radar and sonar at high speed, making timing and prediction are key. The Gun range = Sonar radius. The shells deals more damage the farther they travels. This simulates kinetic charge buildup and encourages long-range engagements.

The Gun has 3 barrels,. Up to three shells can be loaded at once. Each shell takes time to arm, and will fire one at a time per the Captain's discretion. There is no delay in firing loaded shells. The captain can spam click thrice if they want. Ammo is limited and must be used carefully.

Firing Procedure

1. **Enter the Viewfinder:** Switch to the Gun's targeting interface. This is a fixed perspective view aligned with the barrel. It behaves like a scope, showing what the Gun sees: a straight path out of the hull along the starboard line. Multiple targets may appear but the gun will pick the target that is closer in 1) range and 2) to the centre of the gun line.
2. **Arm the Weapon:** Press the ARM button to begin the arming cycle. Shells begin loading into each of the three barrels. You may fire as soon as at least one barrel is loaded.
3. **Maneuver to Align:** The Gun does not rotate. You must fly the ship to bring targets into the firing path.
4. T**arget Lock**: When a valid target enters the viewfinder, the targeting computer will attempt a lock-on to the closest target; this is basically assistive aim. The lock-on takes time and can be interrupted if the target moves out of arc or range
5. **Fire**: Firing without a lock is allowed but less accurate. The shell will fire immediately and travel under raw physics - straight, at high velocity. Misses are common without careful setup.

Broadside Cannons

This is how damage works

Base Damage	10
Damage Mod	1%
Per n Unit Travelled	10
Damage Per Unit	0.01
Distance to Target	10000
Damage to Target	100

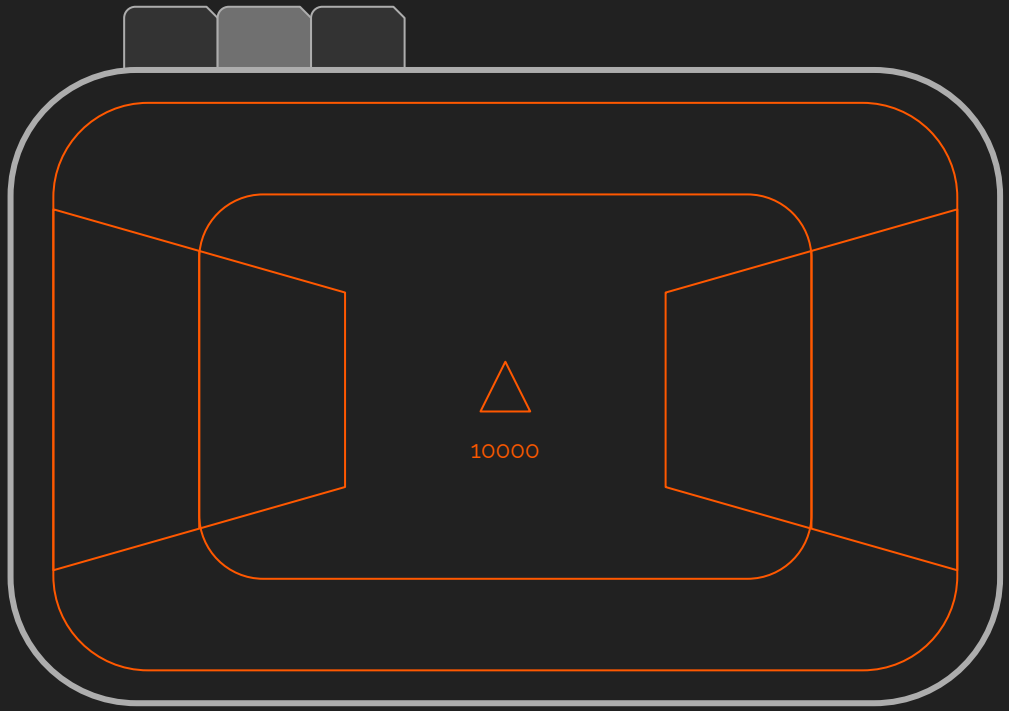
So there is base damage of 10. For every 10 units travelled, the damage increased by 1%. So at 10000 units, the damage from the gun would be 100.

The math is:

DMGPer Unit = (BaseDMG x DMGMod)/Per Unit Travelled) = (10*1%)/10

At distance, the DMG would be **Distance** * **Damage per Unit**. To preserve a base DMG, we will have an IF statement

IF (Distance * DMG per Unit > BaseDMG, Distance * DMG per Unit, BaseDMG)



RAILGUN

The railgun is long distance sniper weapon.

It's kinetic weapon which incredible range and high penetration, with the downside that it takes time to charge, reload and the targeting window is a narrow.

On the flipside, whatever gets hit usually does not get back up.

Firing the gun needs a dedicated targeting computer. The ship needs to come to a full stop, and power needs to be rerouted to begin the sequence.

The gun is best fired for enemies only visible on radar but it can be used in close quarters.

Once fired, power is vented and the ship needs time to recalibrate.

It will also generate a strong energy spike on the Radar, leaving the shooter exposed.

The railgun builds damage over the distance it travels. Only one shot can be fired at the time.



Boarding PODs

Boarding Pods are used to deliver troops onto enemy ships. Much like the Gun, pods are loaded into launch tubes and fired when conditions are met. The system shows a launch timer and chance to hit per pod in the Interaction Panel.

To launch successfully, the enemy ship must be within boarding range and the Star Wolf must be aligned with a side where the target's hull has been breached. The pods will automatically begin locking if these conditions are met and maintained for a few seconds. The player can manually position using the sonar or engage Autopilot for alignment. When ready, the player may hit launch. All pods are launched as one.

Once launched, the panel shows a simplified tactical map of the enemy ship. Troops move autonomously through rooms. The player cannot control them directly but may mark priority rooms and choose between 2 objectives: *Destroy* or *Capture*.

If radar contact with the enemy ship is lost troops will continue their actions without live updates. Once contact is re-established, the player will receive the results.

Troops have individual stats: Health, Damage, and Attack Speed. Surviving troops must be retrieved but coming within range of the enemy ship and pressing Retrieve. Retrieve can also be used as a retreat if the fight is going badly or the player's focus changes.

Successful boarding and defeating enemy ships gives you **comms fragments** which let you decrypt new messages.



PDCs

The PDC is a static black background with four numbered wireframe quadrants, marked 1 to 4. These represent the four active firing arcs the ship can track and target at once. The player does not manually aim; instead, they confirm which threat to engage within each quadrant. The PDC auto-fires once a tag is confirmed.

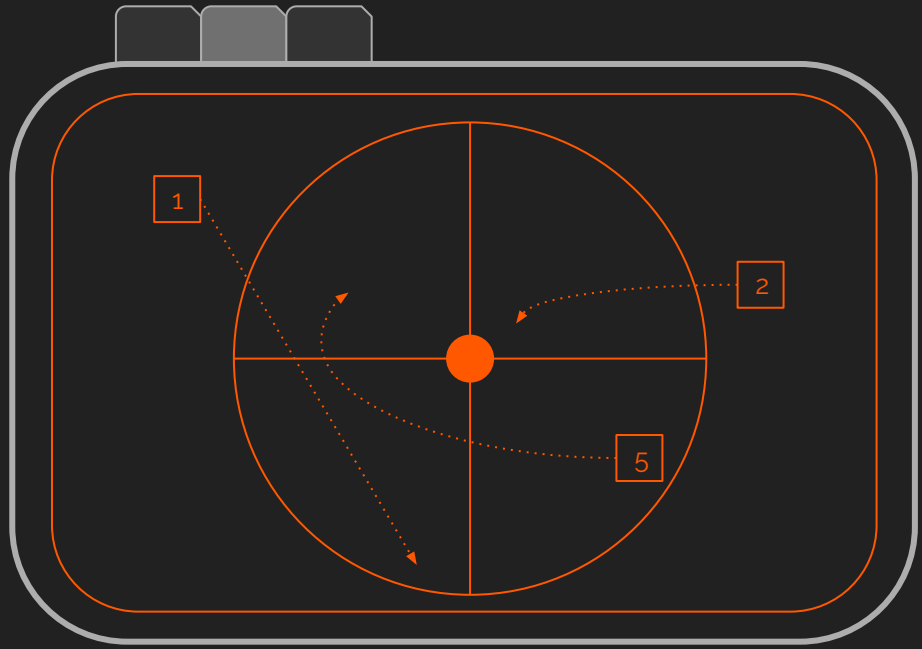
As enemy missiles or fighters enter PDC range, they are assigned to one of the four quadrants. Tags appear in any order and are marked with their corresponding number. Once on screen, each tag pulses for a short time (1.5 to 3 seconds). The player must press the matching number key to confirm the shot before the tag expires.

Pressing the correct key immediately fires on the target,

If the player fails to react in time, the tag expires - indicating a successful hit or escape by the hostile.

Only one target can be engaged per quadrant at a time. If multiple hostiles arrive in the same quadrant, the player may pick a target and wait or tank some hits before that quadrant is free again. Despite this, new threats will continue to appear, and the player can engage any available quadrant simultaneously.

Objects have **health** and PDCs have a **DPS**



MISSILES

To fire missiles you need to have successful scanned the enemy till 100% and then let the systems perform a lock when missiles are active.

Missiles **must target a ship system**

You can select multiple targets with missiles by pressing TAB to toggle between targets.

You have a limited number of missiles.

ABILITIES

Abilities are a set of support and utility functions. There is no creen. These are physical switches which trigger when pressed. There are 5 abilities in the game. 4 have cooldowns and 1 is single-use.

Repair: Dispatches crew to repair hull and one damaged internal system. Life Support and Reactor are prioritised. Automatic. Cooldown: 90 seconds.

Electronic Countermeasures: Breaks all active scanner locks, weapon locks, and blocks new locks for 10 seconds. Missiles in flight will stop tracking you and move in straight lines. Cooldown: 75 seconds.

Signal Ghost: Generates a false telemetry trail (ghost ship) while making you invisible to radar and scanner for 20 seconds. Cooldown: 90 seconds.

Emergency Vent: Activates vent protocols and reactor damping, rapidly reducing power output to 1 and halting all active emissions for 10 seconds. Radar signature drops to near-zero. Cooldown: 120 seconds.

Black Alert (Single Use Only): All systems are maxed to 5 power. Gun and Missiles auto-load and may be fired repeatedly without delay. Gun damage modifier increases to 30%. PDCs auto-engage all threats. Scanner does perfect autoscans of all targets in range. Lasts 30 seconds, after which the ship suffers 2 internal system hits and drops to 0 power for 10 seconds.

COMBAT SYSTEM

So what happens when you land a hit on another ship.

Ships in Starwolf do not have shields. Instead, they have thick plates of armour around the key systems and chambers of the ship.

When a side gets hit, the armour value for that side reduces until it hits **zero**. After this, the inner ship compartments is exposed and subject to the terrible conditions of space - cold, oxygen loss, cosmic radiation to name a few. .

Think of each ship as a **rectangle**. Each side has an armour value or **HULL**. One may typically balance the armour values per side by thinking of **HULL** as a whole stat ratioed between the 4 sides.

In mathematical terms

- Hull(Side) - Weapon DMG = CurrentHull (Side)

But this is too simple and needs more game depth.

Different weapons interact differently with Armour. We can think of this as a damage **TYPE** and penetration ability (**PEN**). **TYPE** looks at the fundamentally effectiveness of a weapon against armour. **PEN** looks at how easily a weapon can pass through something.
The battlefields of Starwolf are littered with the debris of past battles and sometimes you might be shooting through an asteroid or the hulk of a dead ship. PEN is what decides what happens when something is in the way.

For eg. Cannons have the **KINETIC** Type damage, which means they are a bit weaker against **HULL**. Lasers on the other hand are great at burn through **HULL** and penetrating through.

COMBAT SYSTEM - DMG TYPES

Type	Effect	Pen	Description
Kinetic	90% DMG	2	Kinetic weapons are 90% effective against armour and do not really pass through it.
Laser	120% DMG	4	Lasers burn through armour and can get to key systems easily
Plasma	100%, Erode Hull	3	Plasma is average against armour but has a damage over time effect
N-Ray	0% Ignore Hull	1	Radiation has no impact on armour and instead capitalises on existing damage to destroy internals

We will also look at **Keywords**, which add additional rules and conditions. For now we only have a few Keywords

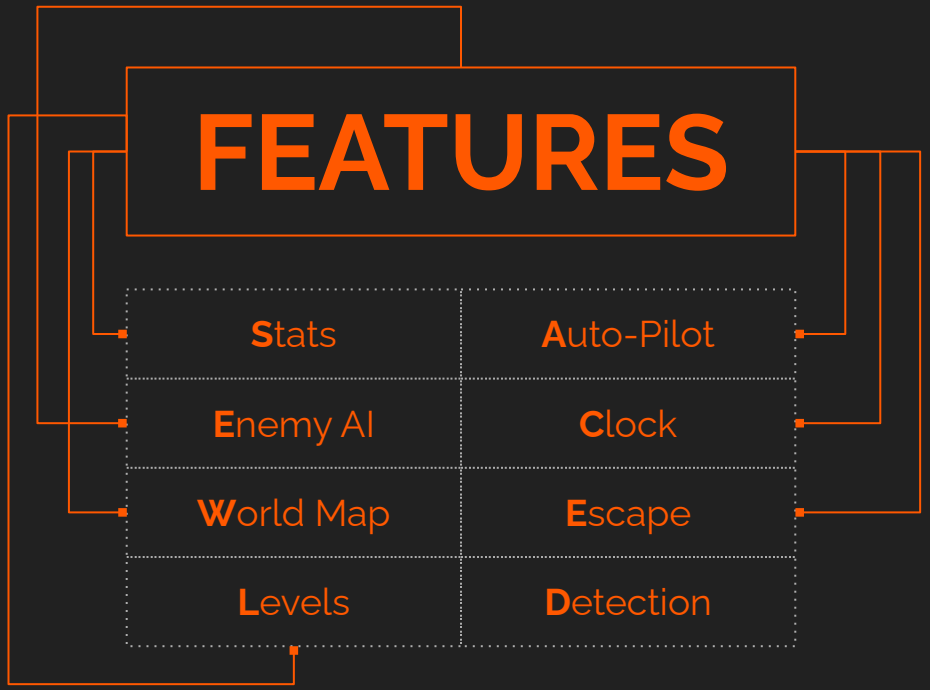
Charge: This primarily applies to the railgun, which is a kinetic weapon. It's PEN stat increases as it charges.
Erode: Erode is a damage-over-time effect which lingers over the hit target, doing additional damage. It refreshes and stacks the more hits you get.
Ignore: ignore the stat completely

COMBAT SYSTEM - PENETRATION

So how does penetration work exactly. To keep it simple, this is a stat comparison - Weapon **PEN** vs Hull **Toughness (TOUGH).**

Pen needs to beat **Toughness** to pass through the obstacle. The greater the difference, the further a projectile will travel after having passed through an object, without losing damage capability. **Both have a maximum rating of 5.**

Condition	Pass	Loss
PEN - TOUGH= x < 1	N	-
PEN = TOUGH	Y	-50% DMG
PEN - TOUGH = x > 1	Y	-50% DMG
PEN - TOUGH = x > 2	Y	-25%DMG
PEN - TOUGH = x > 3	Y	-10%DMG


FEATURES
Stats
Auto-Pilot
Enemy AI
Clock
World Map
Escape
Levels
Detection

STATS

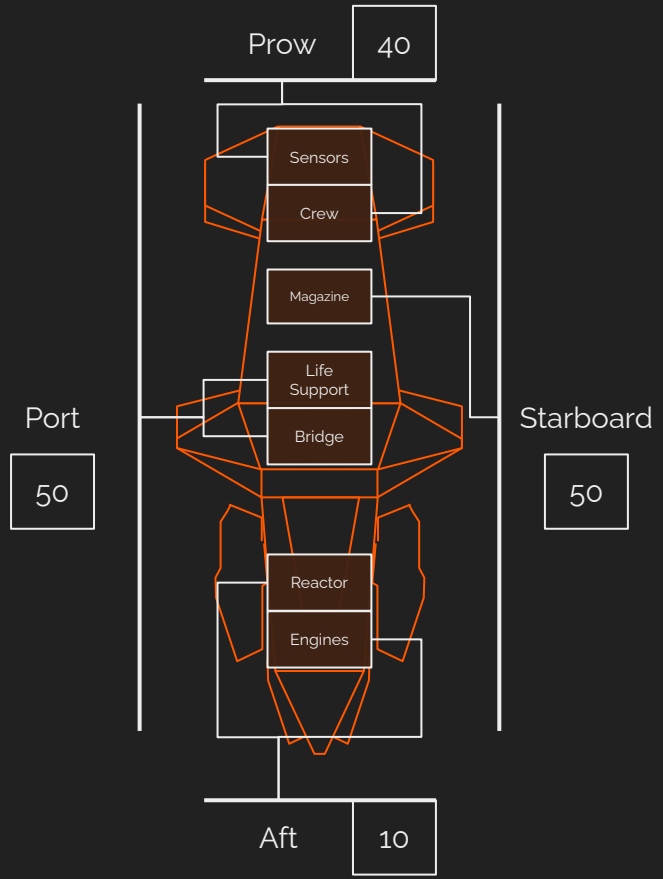
The ship's Hull is divided into four directional quadrants: Port (Left), Starboard (Right), Prow (Front), and Aft (Rear). Each side has its own Health, representing structural integrity in that direction. Incoming damage affects only the side it hits. Once the Hull is breached in a direction, internal components are exposed. Internals include:

- **Bridge** – Governs ship movement and access to the navigation interface. Damaged Bridge causes movement lag or loss of steering.
- **Magazine** – Stores ammo, if hit then damages gun, missiles. PDCs, PODs.
- **Life Support** – Activates a Game Over timer unless repairs are made.
- **Engines** – Reduces max speed, increases turn time.
- **Reactor** – Limits total power. Drains power from random active system
- **Sensors** – Disables scanner, reduces radar range
- **Crew** – Disables Abilities

Each system has its own threshold. If damaged, its related gameplay element is impaired or becomes unavailable until repaired. Each area of the ship is associated with a side.

- Bridge and Sensors tied to Prow
- Reactor and Engines tied to Aft
  Magazine to Starboard
- Life Support and Crew to Port

Eg. If damage to Prow occurs after breach, randomly hit Bridge or Sensors.



AUTO-PILOT

Autopilot allows the ship to orient and travel toward a selected destination without manual steering, but still requires power to Engines and Navigation. The player activates it from the World Map by selecting a coordinate, beacon, signal source, or ship contact. The ship then turns and accelerates based on current turn speed and engine thrust.

Course adjustments follow standard ship handling—if the target is behind, the vessel must rotate before burning. The system does not avoid obstacles, interceptors, or hazards and maintains a direct path unless interrupted. It does not decelerate automatically; the player must cut thrust or take manual control to stop.

Autopilot disengages instantly if the player inputs any movement (e.g. WASD or mouse click).

Combat

During combat, the player can assign a target and configure approach parameters:

- Set Target – Select a ship or object currently visible on radar or scanner.
- Set Range – Choose a preferred engagement distance from predefined bands: 0, 5000, 10000, 18000, or 25000 units.
- Set Orientation – Specify which side to face the target: Port or Starboard.

Once set, the ship will rotate and maintain course and distance accordingly. Autopilot can be interrupted at any time via manual controls. These same parameters form the foundation of AI movement and engagement logic.

WORLD MAP

This is a map of the entire play area.

The shape of the level can be a standard square. The playable area within that can be more distorted shape. Players need to start within the playable area at all times. If they leave, a warning and timer appear.

If the player does not turn back in time, the player is charged with cowardice and the ship self-destructs. The game ends and the player must restart the mission.

The world map slow populates with information as the player explores the level.





LEVEL

CLOCK & PAUSE

The game clock is a universal time tracker used across all systems. It governs enemy ship movement, ETA calculations, scanner sweeps, and mission timing. All intercepted messages reference the game clock to report arrival times, movement speeds, and rendezvous points. Players can use this data - combined with known positions and bearing - to backtrack routes, predict locations, or intercept enemy groups. The clock does not pause, even when systems are offline or the player is in a menu.

This is also used by the game to set up enemy patrols and positions.

The player may pause the game at anytime to make strategic decisions and plan. The clock will not run during this time.

QUESTS

Quests are structured as simple objective chains that track player progress through each mission. Each quest consists of stages like:

1. Locate Target – Triggered when the player enters the correct region or scans a key signal (e.g. identifying the flagship).

2. Eliminate Target – Activated once the flagship or enemy of interest is confirmed on radar. Completed when the ship is destroyed.

3. Exit Region – Requires the player to reach the hyperspace beacon and jump out successfully.

The quest system runs in the background, updating based on in-game triggers like scanner results, ship kills, or reaching coordinates. There are no UI popups or quest journals. This is purely a logic layer for mission flow.

Once you have successfully killed the enemy ship, You must escape from level. A marker will appear on some edge of the world map. You must make your way there to reach the Jump Point and escape.

ENEMY AI


VISUALS

ART STYLE

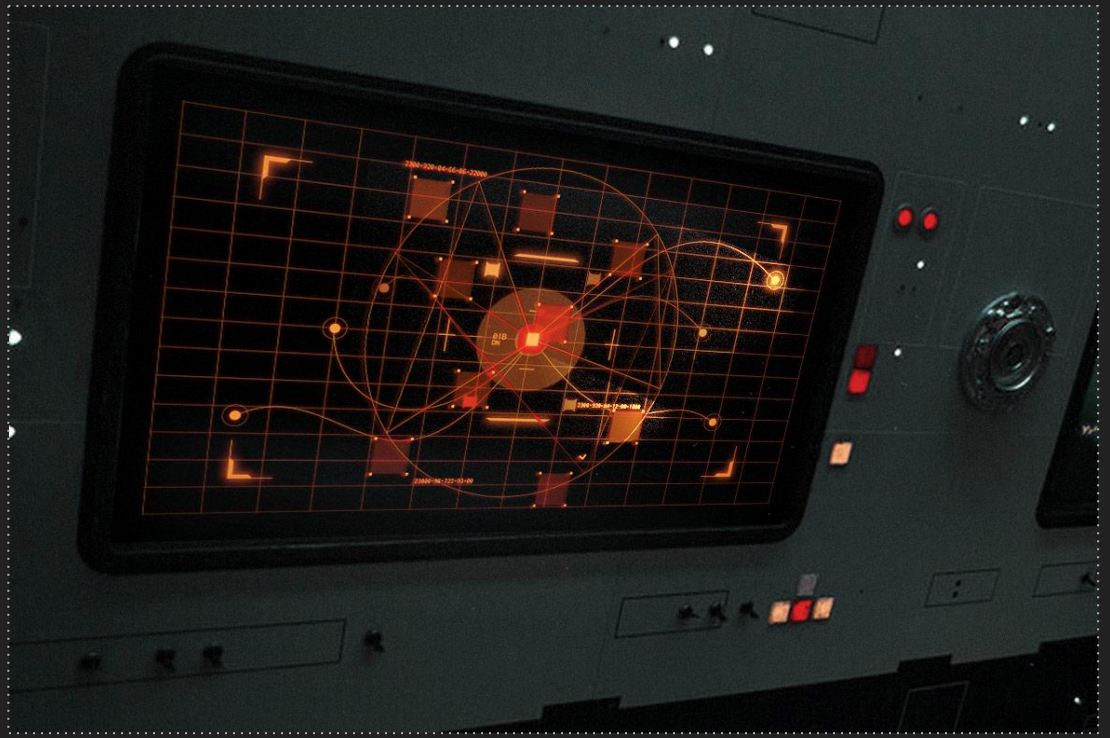
Drawing inspiration from science fiction book covers of the 70's and 80's, Star Wolf's goal is to set itself apart visually from the plethora of space games while preserving the cool edginess this style

Star Wolf Art Direction Pillars;

1. **Retro-Futurism Advanced:** Blending the distinct look of retro sci-fi with the grounded coolness of contemporary technology.

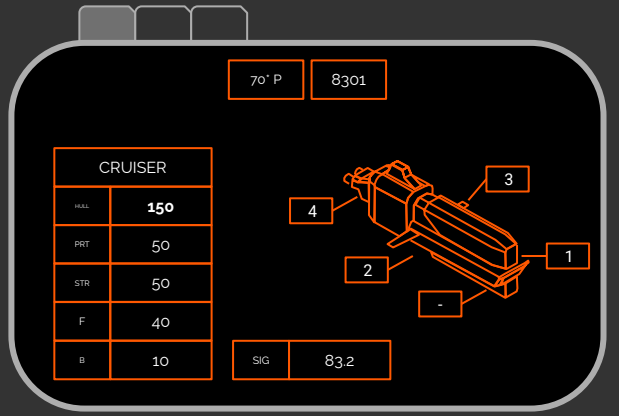
2. **Realistic & Abstract:** A realistic minimalistic terminal driven experience that sets us apart from the hyper-realistic 3D look most sci-fi games aim for.

3. **Visual Variety:** Wide range of player ships, assortment of enemy units and an assortment of environments that speak to the vastness of space, and add depth and personality to the world on the console

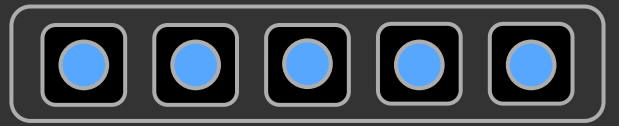


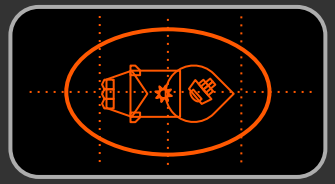





70° P
8301
CRUISER
HULL
150
PRT
50
STR
50
F
40
B
10
SIG
83.2
3
4
1
2
-

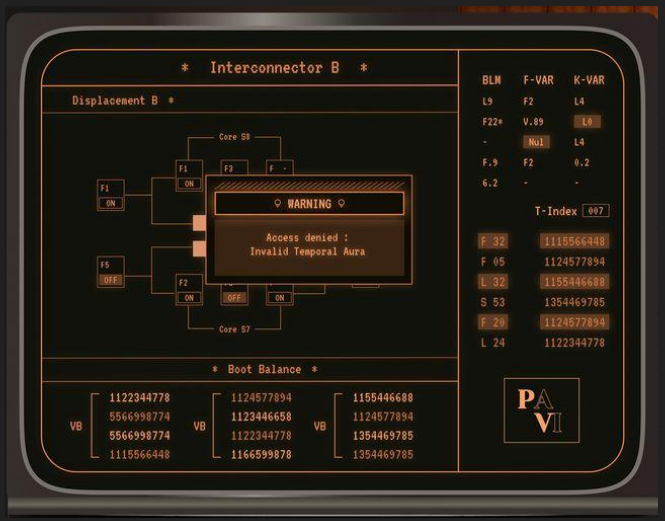

ENG
ARMS
BAY
SPPRT
SIG






100 km/s



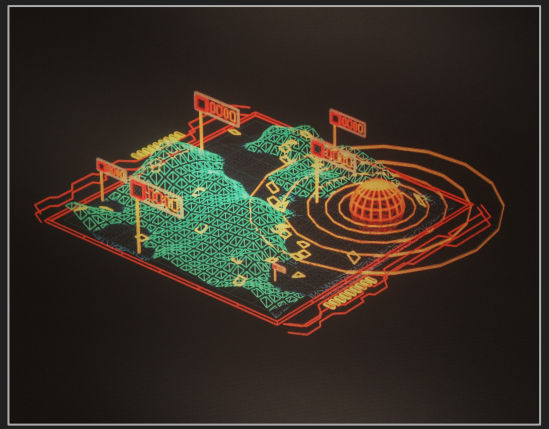


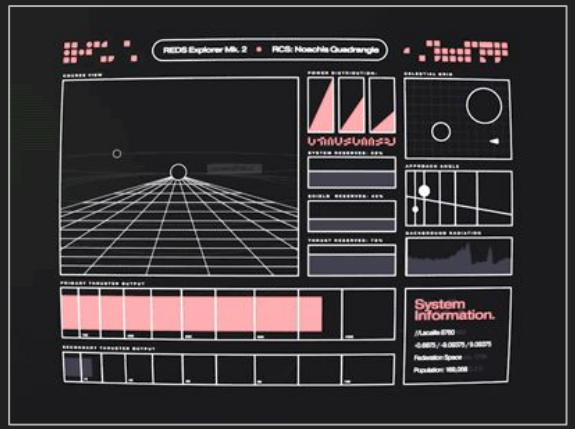


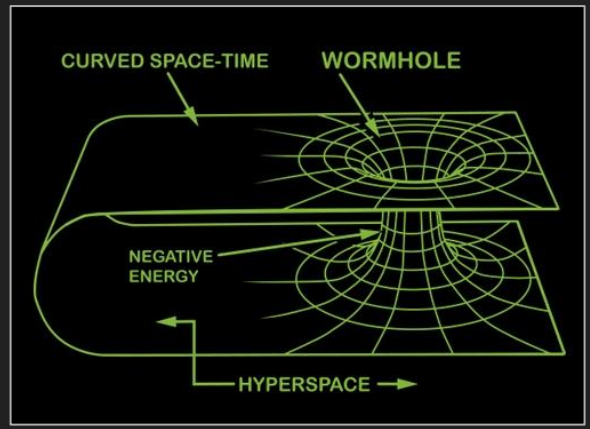
Monochrome, military style layout.
The UI screens are a single colour against black, geometric shapes.
We're going for a slightly retro look so each screen will be set into some kind of metal frame.

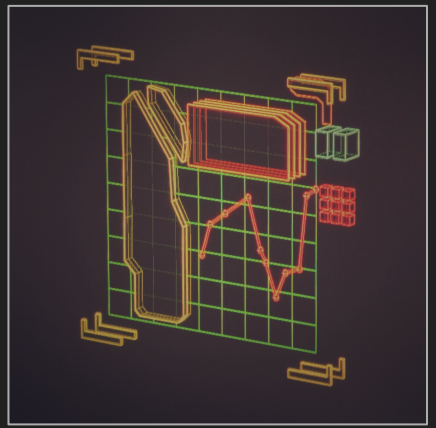
There will be lights and switches.

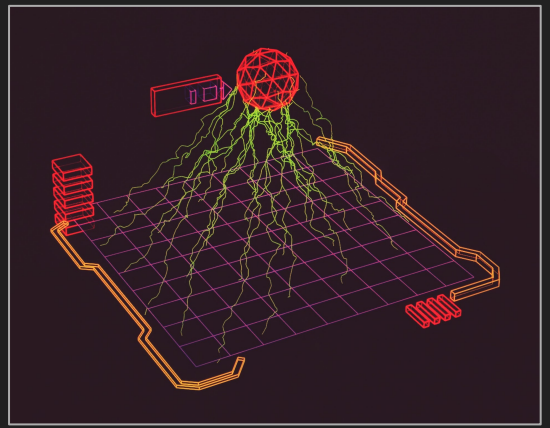
UI Screens

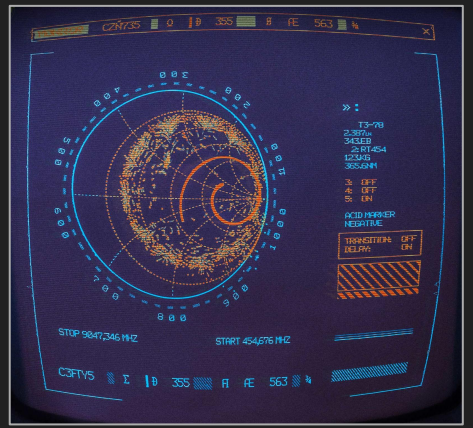


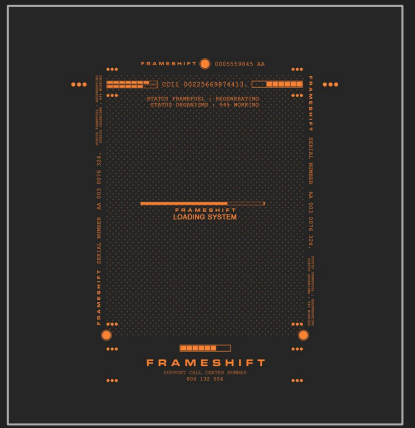

REDS Explorer Mk. 2
RCS: Noachis Quadrangle
System Information.


CURVED SPACE-TIME
WORMHOLE
NEGATIVE ENERGY
HYPERSPACE

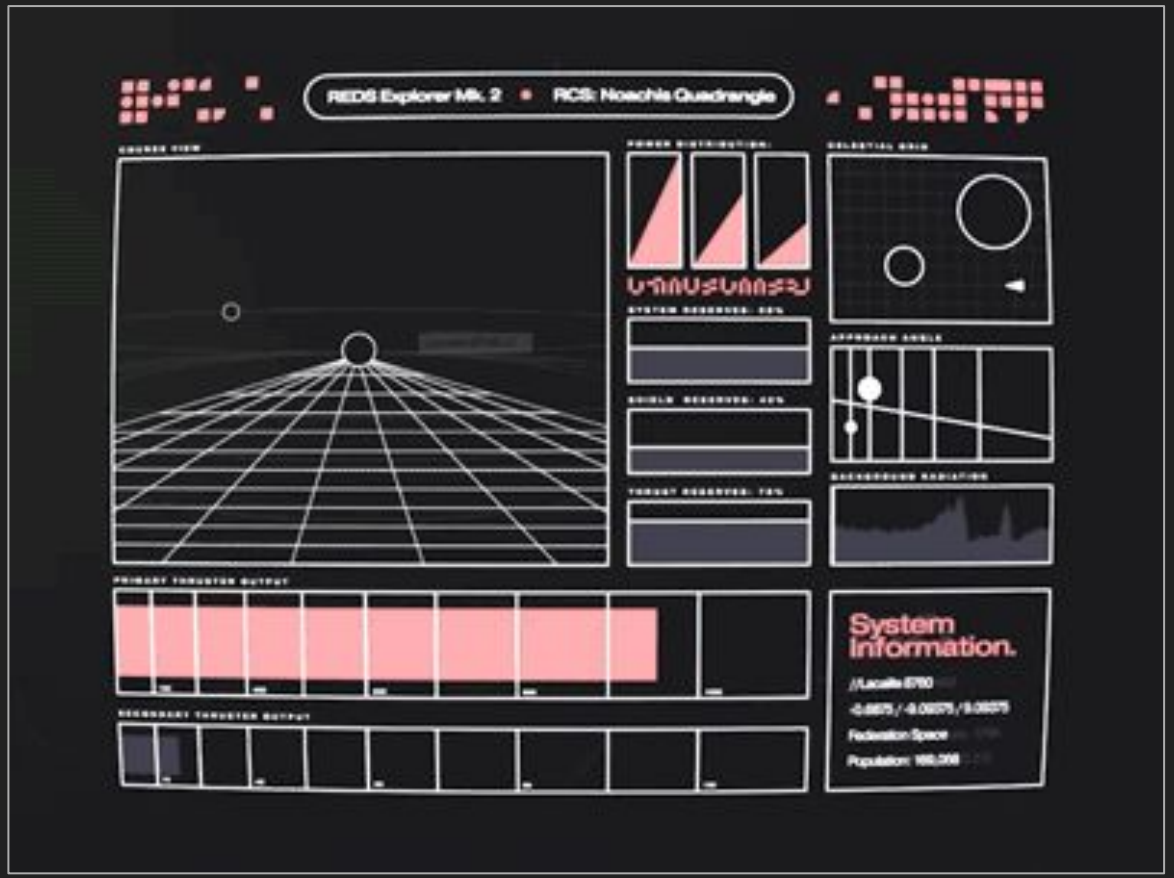





STOP 9847.346 MHZ
START 454.676 MHZ


FRAMESHIFT
LOADING SYSTEM
FRAMESHIFT

UI Screens


REDS Explorer Mk. 2
RCS: Noachis Quadrangle
System Information.

Ship Design

The *Star Wolf* is a wedge-shaped stealth destroyer, designed with a retro-futurist military aesthetic. The wireframe design should emphasise sharp angles, mass, and silhouette clarity. Thematic, we are going for brutal, enduring, rugged, inexorable.

- Wedge-shaped wings with a broad, blunt profile. These are heavy structural elements carrying broadside weapons.
- Hammerhead prow, wide and flat. Strong frontal presence. Looks like a ramming or sensor array section.
- Triangular aft structure enclosing the main thrusters. Compact and well-protected.
- Dorsal tower sits slightly behind the centre, angled forward like a submarine conning tower. Short, sharp, angular.
- Broadsword silhouette from top-down view - triangular blade with midline symmetry.
- Hull structure is layered and angular. Avoid curves. Emphasise panel seams, plating, and embedded turret mounts.
- Avoid ornamental or fragile-looking elements. Prioritise strong, readable shapes that communicate function and survivability.



Ship Design





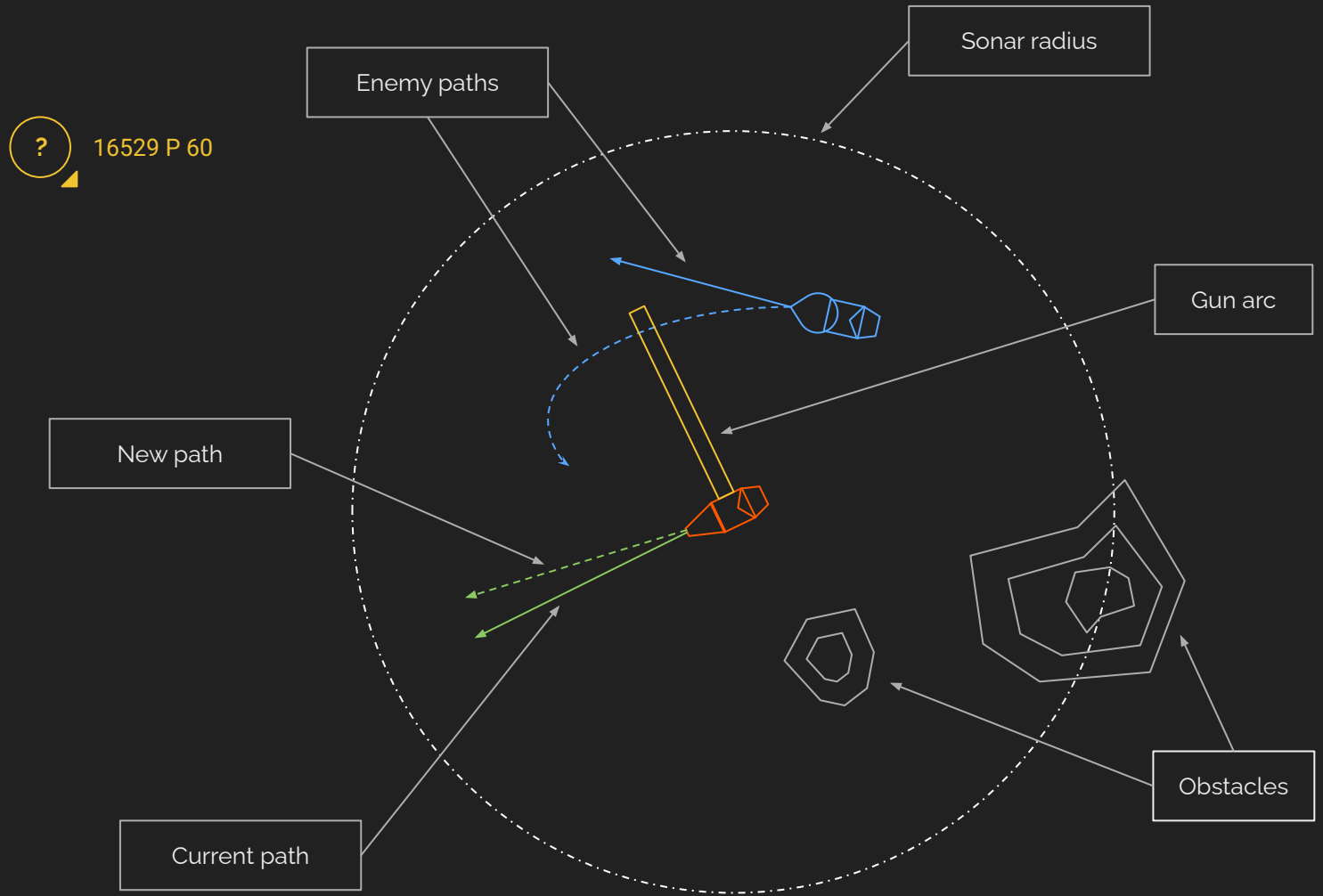
Art Req

1. Concept / Mock paint of Interface
2. Mission Cards
3. Starship Wireframe
4. Interface
5. Size of Ship - 256x256
6.

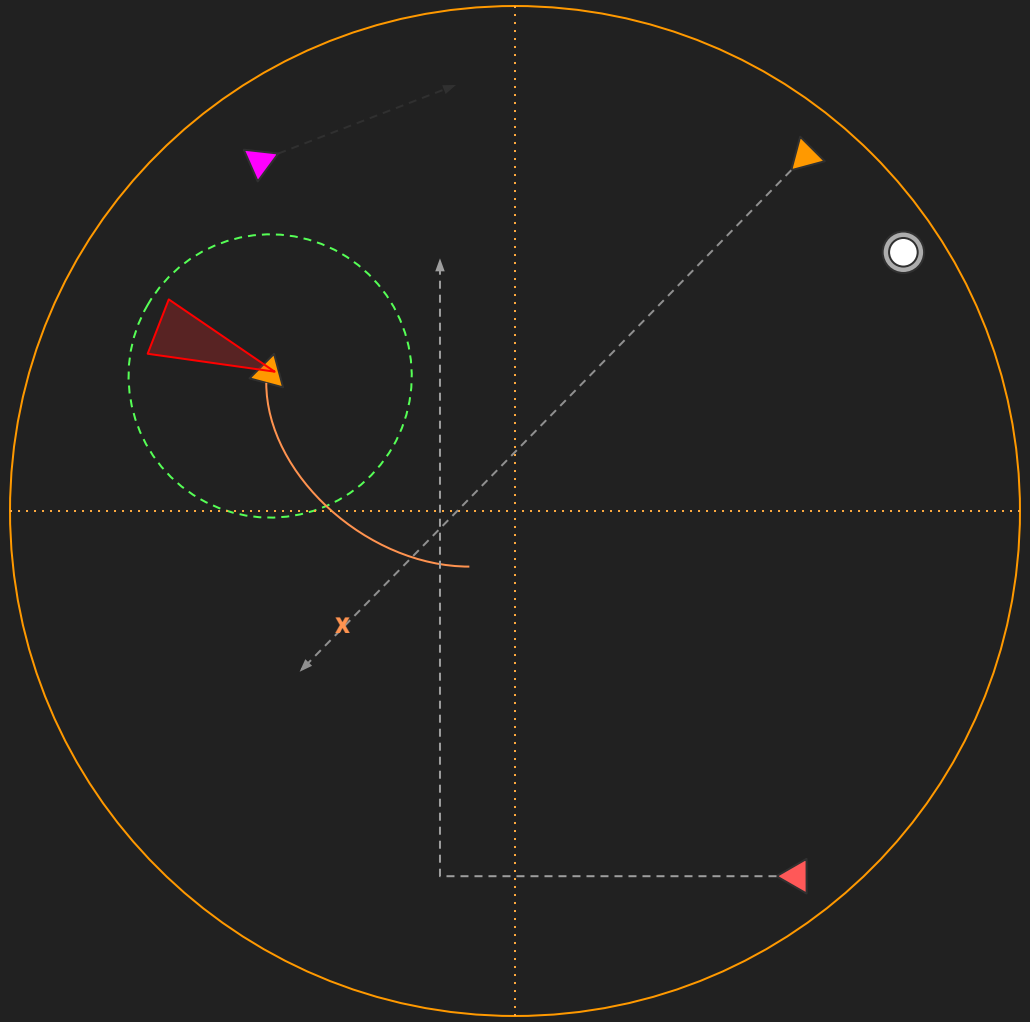
STAR
WOLF

Following slides are notes and **unfinished** concepts

Not part of spec


?
16529 P 60
Enemy paths
Sonar radius
Gun arc
New path
Current path
Obstacles

World Map



- This game is about flying a big spaceship and trying to destroy another big spaceship.
- You are going to fly this ship around a 3D space, which will be rendered on 2d screens in a top down format..
- The fun of the game is that you have to use your instruments
- In this game, you have to manage the position of your spaceship and get it into range and within the angle of fire of your weapons

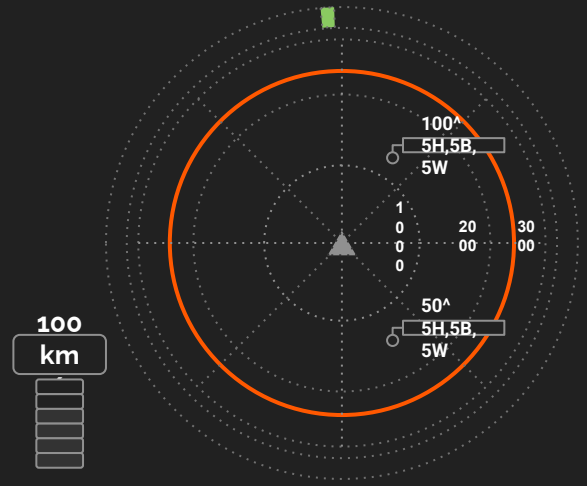
RULES

- The game takes place in real time. Results take time because the distances are very long. Weapons take time to charge. It takes time to travel.
- You win by successfully shooting the enemy ship.
- You lose if you:
    - Die in battle
    - Desert the battlespace
    - Run out of time

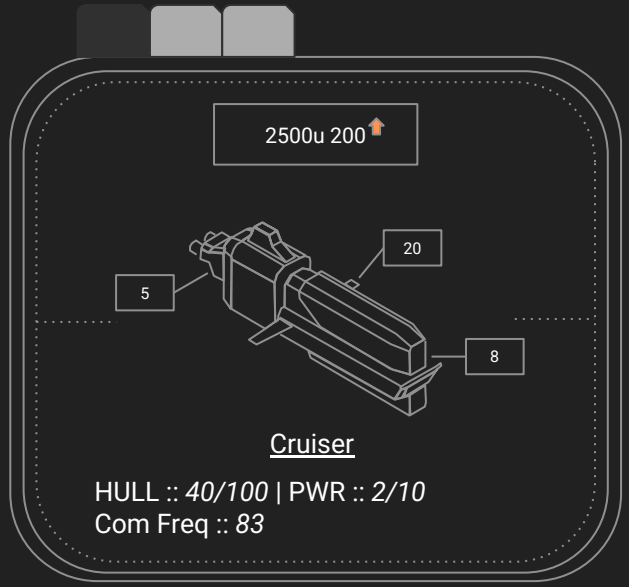


20%
15%
10%
Engines
Weapons
Sensors
Reactor

Navigation



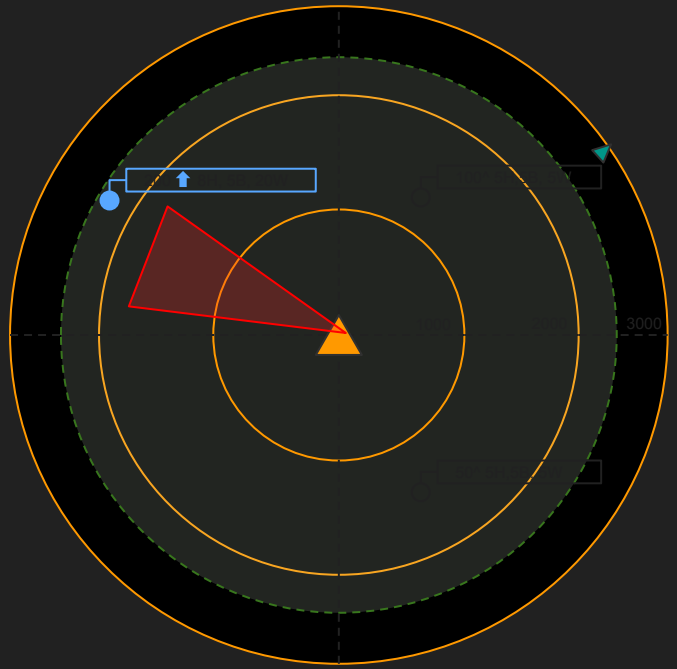
Interactions




ENG
ARMS
BAY
SPPRT
SIG

Navigation Panel: Nav Map + World Map View

Eg. Radius of screen = 3000



In the Navigation view, the player sees a portion of the map **centred around the ship.** The radius of this is adjustable.

When you toggle the world map, you see where the ship is **relative to the centre of the arena**

